

Sharif University of Technology

Department of Mechanical Engineering

Master of Science Thesis

Physics-Informed Modeling of Fluid–Particle Transport in Channels

From Dilute Particle Tracking to Gaussian Parcels and Constrictions

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Abstract

Predicting particle transport requires an accurate carrier flow, a consistent interphase force model, and an evaluation that distinguishes field agreement from reliable particle motion. I develop a staged physics-informed framework that begins with dilute particle tracking in a curved channel, introduces Gaussian parcel coupling in a straight channel, and formulates an extension to a smoothly constricted channel. The baseline combines a continuous velocity–pressure network with a neural approximation of the prescribed Stokes-drag law. Independent finite-element and analytical comparisons give sample-relative errors of 1.510% in curved-channel velocity and 0.863% in gauge-aligned pressure. The normalized drag error is 0.717%. The Gaussian formulation uses the same normalized kernel for fluid interpolation and reaction-force spreading, yielding exact discrete force and work identities on the transfer quadrature. Reduced-model tests recover Stokes relaxation, two-way momentum exchange, Gaussian overlap, and Taylor–Green decay. For these tests, the reported relative errors are below 0.25%; this bound concerns the listed solution observables, not every residual or every channel prediction. Four coupled Gaussian channel approximations are computed and evaluated separately. The selected case has a momentum RMS of 0.0407 against a 0.03 screen, while its parcel kinematic and acceleration RMS values are below 0.002. Baseline trajectory accuracy varies with initial condition: three of seven paths meet all position, time-coverage and endpoint criteria. The final stage develops a proposed constriction study with consistent wall treatment and a reference-based evaluation protocol. The central contribution is a verified foundation for progressing from one-way particle tracking to conservative regularized coupling, together with explicit conditions for extending that foundation to confined geometries.

Keywords: physics-informed neural networks; particle transport; Stokes drag; Gaussian parcels; two-way coupling; channel constriction; numerical verification.

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Abbreviations and Nomenclature

Abbreviation	Meaning
AD	Automatic differentiation
CFD	Computational fluid dynamics
ERPP	Exact regularized point-particle method
FEM	Finite-element method
MLP	Multilayer perceptron
ODE / PDE	Ordinary / partial differential equation
PINN	Physics-informed neural network
RMS	Root-mean-square

Symbol	Meaning
$\boldsymbol{u}, p$	Dimensionless fluid velocity and pressure
$\boldsymbol{q}_i, \boldsymbol{v}_i$	Parcel center and parcel velocity
$\boldsymbol{U}_i$	Fluid velocity interpolated to parcel i
$\boldsymbol{F}_i, \boldsymbol{S}_a$	Represented hydrodynamic drag and reaction source
L_0, U_0, H, B	Reference length, speed, channel height and spanwise depth
$b = B/H$	Reference-depth ratio for planar sphere parcels
d_p, δ_p	Particle diameter scaled by L_0 or H, respectively
$R = \rho_p/\rho_f$	Particle-to-fluid density ratio
$\mathrm{Re}, \mathrm{Re}_p, \mathrm{St}$	Flow Reynolds, slip Reynolds and Stokes numbers
β, α	Mass loading and effective local solid volume fraction
N_i, m_i	Parcel multiplicity and represented dimensionless mass
σ, W_{ai}, Z_i	Gaussian width, normalized transfer weight and normalizer
ΔA_a	Transfer-quadrature area associated with point a
s, η	Centerline coordinate and normalized transverse coordinate
Q, h, H_t	Volume flux per reference depth, half-height and throat height
δ, ℓ, x_c	Constriction severity, half-length and center
E_u, E_p	Discrete relative velocity and gauge-aligned pressure errors

Pressure uses a viscous scale in the baseline and an inertial scale in the Gaussian channel. Each

formulation states its scale explicitly. The Gaussian width σ is distinct from both the particle radius and the channel half-height h.

Chapter 1

Introduction

1.1 Motivation

Particle-laden channel flows connect local hydrodynamics with transport outcomes [1, 2]. A particle can follow the carrier flow closely, cross a streamline because of inertia, or approach a wall closely enough to change its final destination. In a narrow passage, these outcomes depend on both the flow field and the force closure. A small average velocity error can coexist with a significant error in a particle's residence time or contact location. Predictive modeling therefore requires more than visually plausible streamlines.

Conventional computational methods provide established ways to discretize the incompressible Navier–Stokes equations and integrate particle motion [3, 4]. Physics-informed neural networks offer a different representation: a differentiable network approximates the solution, while differential equations and boundary conditions constrain its parameters [5–8]. This representation is attractive when smooth field evaluation and derivatives are needed at many moving particle positions. Its utility must nevertheless be established by comparison with an independent reference and by checks that are sensitive to the intended transport quantities.

I organize this study around a progressive increase in physical coupling and geometric confinement. The first stage considers dilute particles whose motion does not alter the carrier flow. The second introduces represented particle populations and a spatially regularized reaction force in a straight channel. The third defines a smooth constriction as the next geometry to be studied. This progression isolates the consequences of adding two-way coupling before introducing a varying channel height.

1.2 Problem statement

The central problem is to determine which parts of a physics-informed fluid–particle model have sufficient numerical support for transport prediction. There are three distinct questions. The first concerns the carrier field: does the predicted velocity and pressure agree with a

solution of the same boundary-value problem? The second concerns the force transfer: does the discrete coupling apply equal and opposite hydrodynamic forces without introducing artificial interphase power? The third concerns particle transport: are trajectories, contact events, and transit times accurate for the geometry under consideration?

These questions are related but are not interchangeable. An algebraically conservative transfer operator does not eliminate the residual error of a trained fluid field. A low drag-network error does not correct an inaccurate near-wall velocity. A successful straight-channel benchmark does not establish performance at a constriction. I therefore treat each claim at the level at which it has been evaluated.

1.3 Aim and objectives

The aim is to establish a physically consistent and numerically assessable framework for physics-informed particle transport in channels. The study has four objectives:

1. Formulate and evaluate a one-way coupled baseline with independent carrier-flow and analytical drag references.
2. Derive a Gaussian parcel coupling that preserves force and interphase work on its transfer quadrature.
3. Evaluate the components of the coupled formulation using independent analytical and numerical verification problems.
4. Specify a constricted-channel extension whose geometry, boundary conditions, and acceptance criteria can be tested without altering the meaning of the earlier benchmarks.

The corresponding research questions are: How accurately does the baseline recover independently computed channel flow? Which conservation properties follow from the Gaussian discretization, and which require additional solution accuracy? What evidence is still needed before the formulation can support predictions in a constricted channel?

1.4 Scope and physical assumptions

The completed channel models describe incompressible Newtonian flow and spherical heavy particles moving in a plane. The baseline uses one-way Stokes drag and a prescribed irreversible wall-sticking rule. The Gaussian model represents spherical-particle populations through regularized force distributions. The Gaussian cloud is a force-transfer representation; it is not a resolved solid sphere and it is not an independently validated description of a dense particle agglomerate.

The two-dimensional carrier does not resolve spanwise motion. No inference about a finite-depth three-dimensional channel follows from an extruded view. Particle rotation,

lubrication, history force, and hydrodynamic interactions of resolved finite-size bodies lie outside the retained baseline [9–11]. The constriction is considered as a static wall geometry rather than a dynamically growing deposit or a biological vessel.

1.5 Study sequence and contributions

Figure 1.1 summarizes the study sequence. The first contribution is a baseline evaluation that combines analytical plane flow, an independently assembled finite-element reference, a held-out drag test, and an explicit distinction between field and trajectory accuracy. The second is a consistent planar sphere-parcel formulation with shared Gaussian interpolation and spreading. The third is a verification chain connecting reduced relaxation problems, normalization identities, and an unsteady incompressible-flow benchmark. The final contribution is a defined constricted-channel study design that preserves the same requirements for reference matching and conservation.

Completed findings are presented for the baseline carrier field and the Gaussian model's verification problems. The coupled Gaussian channel predictions have unresolved residual errors. The constricted-channel formulation is prospective. This division determines the interpretation of the conclusions and prevents a proposed geometry from being presented as an executed calculation.

1.6 Thesis organization

Chapter 2 reviews the modeling choices and the requirements for credible numerical evidence. Chapter 3 defines the common nondimensional and evaluation framework. Chapter 4 develops the baseline particle problem and reports its supported results. Chapter 5 introduces the straight-channel Gaussian formulation and its verification results. Chapter 6 defines the constricted channel and the proposed evaluation protocol. Chapter 7 compares the implications of the stages, and chapter 8 gives the conclusions and priorities for further work. Supplementary derivations and the computational record appear in the appendices.

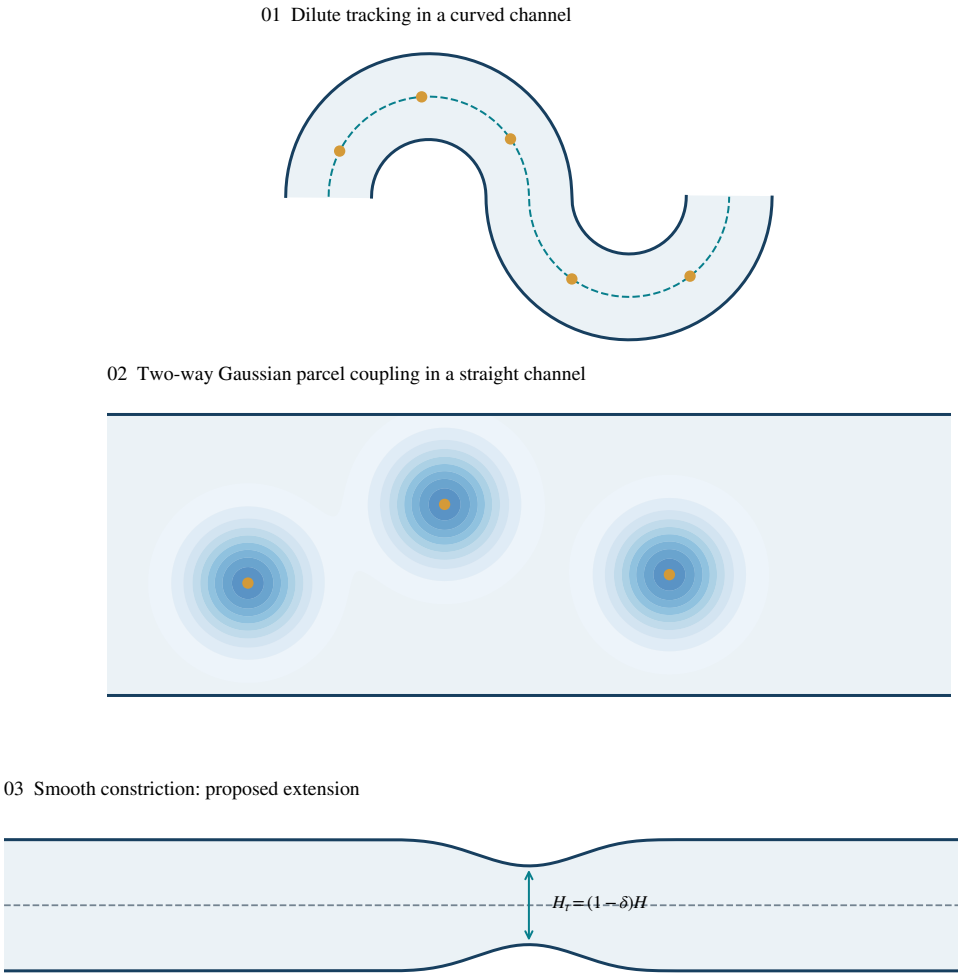


Figure 1.1. Study progression. The first two panels identify the model classes investigated; the third defines the proposed constricted-channel extension. All three panels are geometric illustrations, not simulation fields. Gaussian shading depicts kernel support rather than a measured particle concentration.

Chapter 2

Literature Review

2.1 Modeling levels for particle-laden flow

The hydrodynamics of a particle suspension can be described at different levels of resolution. A particle-resolved method imposes boundary conditions on each particle surface. It can represent the disturbance flow, finite-size effects, and surface stresses when these are sufficiently resolved. Direct forcing immersed-boundary methods provide one established route to such calculations [12, 13]. Their additional resolution requirements are justified when particle-scale flow is itself an essential part of the problem.

A point-particle model instead advances particle centers using a force law obtained under specified physical assumptions [2]. The classical equation of motion for a small rigid sphere includes effects beyond steady drag [9]. The heavy-particle, low-slip-Reynolds-number approximation used here retains a much narrower description. It is appropriate to assess this approximation as a simplified model rather than attribute to it the full physics of the parent equation.

The strength of a point-particle approach is its economy of representation. Its limits follow from the same reduction. A particle center and diameter do not impose no-slip conditions on an actual moving surface. Finite-size wall stresses, rotational dynamics, and lubrication do not appear unless additional closures are included. These distinctions are particularly important when a channel becomes narrow relative to particle diameter [4].

2.2 One-way and two-way coupling

In one-way coupling, the fluid field determines the particle force but the particle reaction is neglected in the fluid equations [1, 14]. This separation is useful for studying dilute transport because the fluid solution can be evaluated independently of particle initial conditions. Small particle volume fraction is a necessary consideration, although it does not alone guarantee negligible feedback when the particle density is high. Mass loading and the relevant time scales also matter [1, 2]. These broad multiphase-flow classifications provide modeling

context; turbulence-based regime maps are not imposed as universal thresholds for this laminar channel.

Two-way coupling includes the hydrodynamic reaction in the carrier-fluid momentum balance. The coupling must specify how a particle samples the fluid and how its reaction is distributed to the Eulerian field. Treating either operation independently can generate an inconsistent power exchange even when the integrated force is correct. A shared transfer operator provides a direct means to avoid this particular inconsistency [15].

Regularized point-particle methods provide a principled context for this transfer. In the exact regularized point-particle method, the treatment of the disturbance and its regularization is tied to the governing flow equations [16]. A generic normalized Gaussian deposition does not inherit all of that method's properties. In particular, sampling the disturbed velocity at a force source is not equivalent to recovering the undisturbed velocity required by an isolated-particle drag law [17, 18].

2.3 Gaussian parcels and represented populations

A parcel represents a population of physical particles through a multiplicity [2]. The planar mass convention used here is specified in equation (3.9). Its represented mass must scale with that multiplicity, and its reaction force must use the same convention. A Gaussian kernel spreads that force over a finite region, avoiding a singular source in the carrier equations. The width of the kernel is a modeling parameter with a direct effect on the spatial distribution of forcing [19, 20]. These coarse-graining and Euler–Lagrange methods supply background for transfer and resolution; their algorithms are distinct from the neural representation used here.

Three lengths must be kept distinct: the physical diameter, the numerical or model regularization width, and the local channel height. Equating the Gaussian width with a particle radius without justification would blur point-particle and finite-size interpretations. Likewise, changing the width while keeping quadrature resolution fixed does not constitute an ordinary grid-convergence study: it changes the regularized model being solved.

Near a wall, a free-space Gaussian is truncated by the fluid domain. A discrete normalization can preserve its integrated weight on the chosen quadrature. That normalization establishes a transfer identity; it does not derive a wall-corrected drag law [10, 11]. This distinction carries directly into the constricted-channel extension developed later.

2.4 Physics-informed representations

Physics-informed neural networks approximate solution fields with differentiable functions and train those functions using equation and boundary residuals [5, 7, 21]. They provide continuous field values without a separate spatial interpolant. Automatic differentiation also

allows the same representation to supply the derivatives entering the equations [22]. These features are useful for particle transport, where the evaluation locations move continuously.

Physics information in the objective is not a guarantee that optimization finds a sufficiently accurate solution [23]. Residual terms can have different scales and different optimization behavior. Gradient imbalance and training difficulties are documented limitations of PINNs [24, 25]. A weighted objective may decrease while a transport-sensitive region remains inaccurate. Consequently, reporting only the optimization history would provide insufficient support for the predictions considered here.

Hard constraints offer an additional design choice. An ansatz can impose an initial condition, periodicity, or homogeneous wall velocity exactly, leaving the network to represent a correction that respects those conditions [26, 27]. This reduces the number of requirements enforced through a penalty. It does not remove the need to evaluate momentum, continuity, and particle-motion residuals at independent points.

The baseline and Gaussian stages use different neural roles. The baseline flow network approximates a steady velocity–pressure field, whereas the force network approximates a prescribed algebraic law. In the Gaussian stage, separate fluid and parcel networks are coupled through common residuals. This is a continuous-time coupled forward representation rather than a trajectory regression against reference data.

2.5 Independent references and numerical credibility

The finite-element method supplies an independently discretized reference for the baseline carrier flow. Taylor–Hood velocity–pressure elements are a standard mixed formulation for incompressible flow [28]. Their assembly here uses `scikit-fem` [29]. An independent discretization is valuable because its numerical errors differ from those of the neural representation. It must still be evaluated for nonlinear convergence, incompressibility, and mesh sensitivity.

Verification and physical validation address different questions [30, 31]. Agreement with an analytical solution tests whether a specified mathematical problem is solved correctly. Agreement with another numerical solver tests consistency for a matched model and boundary-value problem. Experimental validation additionally requires suitable measurements, controlled conditions, and compatible physical assumptions. The results in this thesis primarily concern the first two questions.

A relative velocity error should be interpreted together with local discrepancies, boundary errors, flux variation, and particle-motion diagnostics. Pressure requires particular care because incompressible velocity boundary conditions determine pressure only up to an additive constant. Comparing unaligned pressure arrays can produce a large error without a corresponding physical disagreement. Conversely, an error measured on a section where the

reference pressure is constant can have an ill-conditioned denominator. A spatially extended, gauge-aligned comparison avoids these problems.

2.6 Transport accuracy and confinement

Particle endpoints depend on a time-integrated response to the carrier field. A small discrepancy may accumulate along a long path or determine whether a particle approaches a capture boundary. Near-wall outcomes can therefore be less accurate than bulk flow metrics suggest. A meaningful trajectory comparison must pair the same initial conditions, restrict its position error to the common time interval, and separately report whether the two solutions reach the same endpoint class.

Confinement adds a physical limitation to this numerical sensitivity. When the local channel height approaches the particle diameter, the unbounded Stokes law becomes a questionable description even if the numerical trajectory is well resolved [10, 11]. A study of narrowing should therefore track both numerical agreement and the changing ratio of particle diameter to local gap. A dilute regularized model cannot be used to infer dense blockage merely because a narrow geometry is drawn.

2.7 Position of the present study

Table 2.1 summarizes the modeling levels relevant to the study. The purpose is to build a verified chain across the first two levels and specify the requirements for a constrained extension. The focus is neither a new empirical drag correlation nor a replacement for resolved particle hydrodynamics. The main methodological question is whether a neural representation can support the stated fluid and particle quantities while preserving a consistent transfer of force and work.

Table 2.1. Model classes and the evidence they require.

Model	Main representation	Essential assessment
One-way point particles	Carrier field and particle ODEs	Field error, slip regime, trajectory agreement
Gaussian parcels	Population-weighted regularized forcing	Shared transfer, residuals, budgets, width sensitivity
Resolved particles	Moving solid surfaces and surrounding flow	Surface resolution, forces, torques and gap flow
Constricted dilute model	Variable-height fluid domain with particles	Wall geometry, flux, throat resolution and closure validity

Chapter 3

Mathematical and Evaluation Framework

3.1 Baseline nondimensionalization

Let L_0 and U_0 denote reference length and velocity, and let ρ_f and μ denote the fluid density and dynamic viscosity. The baseline uses

$$\boldsymbol{x}^* = \frac{\boldsymbol{x}}{L_0}, \quad t^* = \frac{tU_0}{L_0}, \quad \boldsymbol{u}^* = \frac{\boldsymbol{u}}{U_0}, \quad p_v^* = \frac{pL_0}{\mu U_0}, \quad \mathrm{Re} = \frac{\rho_f U_0 L_0}{\mu}. \tag{3.1}$$

Stars are suppressed after the scales have been specified. The steady carrier equations are

$$\nabla \cdot \boldsymbol{u} = 0, \qquad \mathrm{Re}(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p_v + \nabla^2 \boldsymbol{u}. \tag{3.2}$$

Using a viscous pressure scale places Re on the convective term. Pressure values from this model cannot be compared directly with a model using an inertial pressure scale without conversion.

For a spherical heavy particle, the retained force is the creeping-flow drag. With $R = \rho_p/\rho_f$ and $d_p = d/L_0$, its dimensionless equations are

$$\frac{d\boldsymbol{q}}{dt} = \boldsymbol{v}, \qquad \frac{d\boldsymbol{v}}{dt} = \frac{18}{R\mathrm{Re}d_p^2}[\boldsymbol{u}(\boldsymbol{q}) - \boldsymbol{v}] = \frac{\boldsymbol{u}(\boldsymbol{q}) - \boldsymbol{v}}{\mathrm{St}}, \qquad \mathrm{St} = \frac{R\mathrm{Re}d_p^2}{18}. \tag{3.3}$$

The slip Reynolds number is

$$\mathrm{Re}_p = \mathrm{Re}d_p|\boldsymbol{u}(\boldsymbol{q}) - \boldsymbol{v}|. \tag{3.4}$$

The physical rationale for Stokes drag requires $\mathrm{Re}_p \ll 1$, along with a small particle relative to the spatial scales of the carrier flow.

3.2 Geometry and boundary-value matching

The baseline channel is a constant-width tube around a planar centerline $\boldsymbol{c}(s)$. Let $\boldsymbol{n}(s)$ denote its unit normal and let h denote the half-width. Coordinates are mapped by

$$\boldsymbol{x}(s,\eta) = \boldsymbol{c}(s) + h\eta\boldsymbol{n}(s), \qquad 0 \le s \le L, \quad -1 \le \eta \le 1. \tag{3.5}$$

The carrier velocity vanishes on both walls. At each end section, a parabolic velocity in the local tangent direction is prescribed:

$$\boldsymbol{u}(s_{\mathrm{end}},\eta) = (1-\eta^2)\boldsymbol{t}(s_{\mathrm{end}}). \tag{3.6}$$

Thus the reference problem uses velocity conditions at both ends, rather than an imposed pressure outlet. A pressure gauge fixes the remaining constant. The neural and finite-element comparisons must use this same problem.

The associated volume flux per unit reference depth is

$$Q_0 = h\int_{-1}^{1}(1-\eta^2)\,d\eta = \frac{4h}{3}. \tag{3.7}$$

For a steady incompressible flow without sources of fluid mass, the section flux should be independent of s. This makes flux variation a useful integrated diagnostic alongside local differential residuals.

3.3 Gaussian-channel scales

For the Gaussian stage, the reference length is the channel height H. The pressure scale is inertial:

$$p_i^* = \frac{p}{\rho_f U_0^2}, \qquad \mathrm{Re}_H = \frac{U_0 H}{\nu}, \qquad \delta_p = \frac{d}{H}. \tag{3.8}$$

For identical length and velocity scales, the two pressure conventions satisfy $p_\nu^* = \mathrm{Re}_H p_i^*$. The governing equation is correspondingly written with $1/\mathrm{Re}_H$ multiplying viscous diffusion.

The carrier is two-dimensional, but the represented physical particles are spheres. A reference depth B is therefore required to convert represented sphere mass into a planar source. With $b = B/H$ and multiplicity N_i, the dimensionless represented mass is

$$m_i = \frac{N_i R\pi\delta_{p,i}^3}{6b}, \qquad \mathrm{St}_i = \frac{R\mathrm{Re}_H\delta_{p,i}^2}{18}, \qquad \beta = \frac{\sum_i m_i}{|\Omega|}. \tag{3.9}$$

This convention scales physical mass by $\rho_f H^2 B$. The kernel has units of inverse area before nondimensionalization. Replacing it with a three-dimensional volume kernel without changing

the mass convention would be dimensionally inconsistent.

3.4 What is learned

The neural representations approximate fields or trajectories; the physical equations specify their constraints. The baseline force network learns a known drag response from an algebraic residual. It does not infer an unknown force law from experiments. The Gaussian parcel network predicts time-dependent position and velocity corrections whose consistency is enforced by both kinematic and dynamic residuals.

Adam is used for initial optimization, with optional limited-memory quasi-Newton refinement [32, 33]. The optimizer sequence does not define the credibility of the final state. Checkpoint selection uses a fixed residual monitor, whereas the reported field comparison uses an independent reference. A longer training budget is a computational choice, not an accuracy certificate.

3.5 Error measures and pressure gauge

For values at M independent locations, the sample-relative velocity error is

$$E_u = \left[\frac{\sum_{j=1}^{M} |\boldsymbol{u}_j^{\mathrm{PINN}} - \boldsymbol{u}_j^{\mathrm{ref}}|^2}{\sum_{j=1}^{M} |\boldsymbol{u}_j^{\mathrm{ref}}|^2} \right]^{1/2}. \tag{3.10}$$

These are discrete norms over the specified samples. Unless area weights are explicitly included, they are not exact domain-integral norms. For pressure, I first remove the arithmetic mean of each compared array:

$$\widetilde{p}_j = p_j - \frac{1}{M}\sum_{k=1}^{M} p_k, \qquad E_p = \frac{\|\widetilde{\boldsymbol{p}}^{\mathrm{PINN}} - \widetilde{\boldsymbol{p}}^{\mathrm{ref}}\|_2}{\|\widetilde{\boldsymbol{p}}^{\mathrm{ref}}\|_2}. \tag{3.11}$$

The pressure samples span the streamwise direction so that the reference denominator is meaningful. Mean subtraction removes only a constant; it does not rescale a pressure gradient or modify the predicted dynamics.

The section-flux diagnostic is

$$E_Q = \max_s \left| \frac{Q(s)}{Q_0} - 1 \right|. \tag{3.12}$$

Reported differential residuals use their stated nondimensional equations. Their magnitudes should not be compared across pressure scales, equation normalizations, or geometries without accounting for those differences.

3.6 Trajectory evaluation

Consider a neural and reference trajectory with the same initial condition. If their common interval is $[0, T_c]$, the position discrepancy is measured over that interval and normalized by channel width $W = 2h$. Endpoint agreement is assessed separately using the terminal event class. I also retain the fraction of the longer trajectory covered by the common interval. A low error calculated over a short common interval is not sufficient evidence of agreement over the full transit.

Integration success has a narrower meaning: it indicates that the numerical ODE solver completed its assigned calculation successfully. It does not establish that the neural velocity or force was accurate. Accordingly, the acceptance criteria distinguish integration status, position discrepancy, common-time coverage, and endpoint agreement.

3.7 Evidence hierarchy

I use four levels of numerical evidence. Algebraic identities establish properties of the coupling operators. Analytical benchmarks test reduced equations with known solutions. Independently discretized references test the field approximation for a matched problem. Finally, transport comparisons assess the accumulated consequences of the field and force errors. None of these alone establishes experimental agreement.

For the baseline, the numerical screening limits are 5% for field and normalized drag errors, 2% for boundary RMS and reference-interpolation error, and 5% for maximum flux deviation. Baseline particle acceptance is stricter in its combination of conditions: every tested endpoint class must agree, every width-normalized RMS position error must be below 5%, and every common-time fraction must exceed 95%. For the Gaussian verification tests, the stated relative-error and independent residual criteria are 0.03. These limits are study criteria rather than universal physical thresholds. Their actual outcomes are reported where the corresponding results are presented.

Chapter 4

Baseline Particle Problem

4.1 Physical problem and parameter set

The baseline separates carrier-flow approximation from particle response. The carrier is steady and two-dimensional. Particles are spherical, heavy, and dilute, and their forces do not feed back into the fluid. The principal channel consists of two connected semicircular bends with nominal inner and outer radii $R_{\text{in}} = 0.2$ and $R_{\text{out}} = 0.5$. Its width is $W = 0.3$ and its centerline length is approximately 2.1991, all scaled by L_0. The Reynolds number is $\text{Re} = 1$.

The baseline particle parameters are $R = 1000$ and $d_p = 0.025$. The resulting Stokes number is $\text{St} \simeq 0.03472$. The channel wall rule stops the particle when its center approaches within one radius of a wall. This is a specified irreversible sticking model, not a prediction of adhesion chemistry or a resolved impact process. In the full trajectory set, the largest recorded slip Reynolds number is approximately 0.01221, which supports the low-slip-Reynolds-number assumption for these calculations. The ratio $d_p/W \simeq 0.0833$ still warrants caution for near-wall motion.

4.2 Flow and force representations

4.2.1 FlowNet

The flow network maps five input slots to velocity and pressure:

$$(x, y, R_{\text{in}}, R_{\text{out}}, \text{Re}) \longmapsto (u, v, p_v). \tag{4.1}$$

Velocity–pressure PINN formulations for incompressible flow provide related methodological background [34]. The present network has five hidden layers with 70 tanh units per layer. The spatial inputs are centered and scaled, and the pressure output is scaled with a channel-based pressure magnitude. The geometry and Reynolds inputs are fixed within an individual trained model. Their presence does not establish generalization across untrained geometries or Reynolds numbers.

The objective combines momentum and continuity residuals, wall and end velocity conditions, a pressure gauge, and a section-flux condition. If $r_c = \nabla \cdot \boldsymbol{u}$ and $\boldsymbol{r}_m$ denotes the residual of equation (3.2), the scaled physics term is

$$\mathscr{L}_{\text{phys}} = \left\langle (h r_c)^2 \right\rangle + \left\langle |h^2 \boldsymbol{r}_m|^2 \right\rangle . \tag{4.2}$$

Here the averages follow the componentwise mean-square convention of the implementation. The complete objective uses weights 20 for the velocity boundary term, 1 for the scaled gauge term, and 5 for the normalized flux term. Finite-element interior values are not used in this objective.

4.2.2 ForceNet

The second network has five hidden layers with 90 tanh units and maps

$$(u, v, p_v, d_p, x_p, y_p, u_p, v_p) \longmapsto (F_x, F_y). \tag{4.3}$$

The force scale is the displaced-fluid mass times U_0^2/L_0. Under this convention, the dimensionless Stokes force and particle acceleration are

$$\boldsymbol{F}^* = \frac{18}{\mathrm{Re} d_p^2}(\boldsymbol{u} - \boldsymbol{v}), \qquad \frac{d\boldsymbol{v}}{dt} = \frac{\boldsymbol{F}^*}{R}. \tag{4.4}$$

The neural approximation is trained on the normalized slip response. Subtracting its evaluation at zero slip from its evaluation at the actual slip enforces exactly zero force when $\boldsymbol{u} = \boldsymbol{v}$. The network therefore approximates a specified drag relation. Its agreement with that relation is an approximation test, not discovery of a new constitutive law.

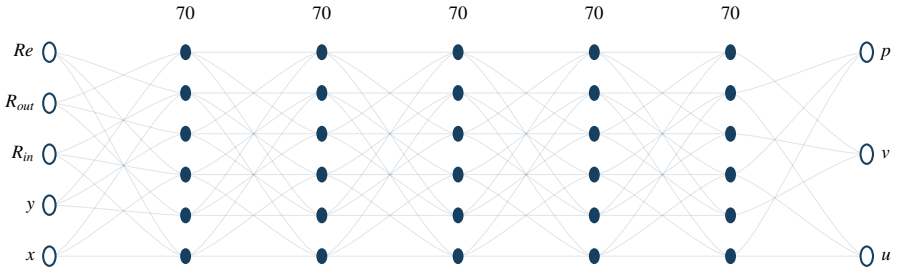
Figure 4.1 gives a visual summary of both networks. The role of ForceNet is intentionally kept separate from the carrier-field approximation so that the two errors can be evaluated independently.

4.3 Independent finite-element reference

The reference solves the same incompressible equations using quadratic velocity and linear pressure Taylor–Hood elements [28, 29]. A Picard iteration treats the convective term, and sparse direct solution is used for the linear subproblems. The finest channel mesh contains 3,026 triangles, 12,530 velocity degrees of freedom, and 1,620 pressure degrees of freedom.

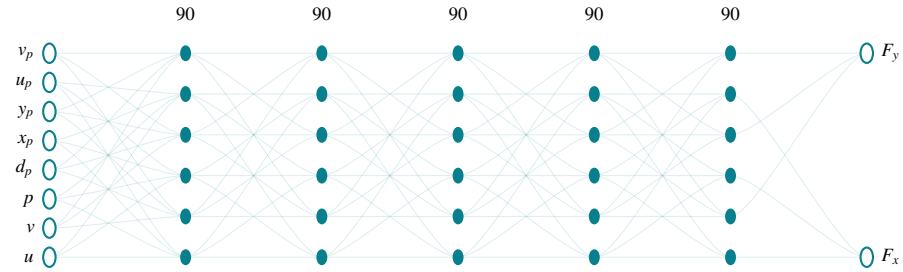
Three meshes assess sensitivity. Table 4.1 shows that the difference between the two finest velocity fields is 0.173%. The finest reference has a nonlinear residual of approximately 1.60×10^{-10}. Its reported divergence norm is not identically zero; the mixed discretization

FlowNet | continuous fluid field



Five hidden layers · tanh activation · PDE and boundary losses

ForceNet | prescribed Stokes response



Five hidden layers · exact zero-slip constraint · algebraic drag loss

Figure 4.1. Baseline neural architecture. Each row represents a fully connected MLP; the few displayed nodes are illustrative, while the numbers above hidden layers give their actual widths. FlowNet supplies velocity and pressure. ForceNet approximates the normalized Stokes response and uses a zero-slip subtraction. The first network is constrained by PDE and boundary residuals; the second by an algebraic drag residual.

enforces incompressibility weakly. The mesh differences are sensitivity indicators, not a formal uncertainty estimate or a grid-convergence index [30, 35].

Table 4.1. Reference-mesh assessment. Velocity differences are measured relative to the finest field on the comparison samples.

Triangles	Velocity DOFs	Pressure DOFs	Difference (%)	Nonlinear residual
290	1298	180	1.141	6.735×10^{-11}
1298	5474	720	0.173	1.097×10^{-10}
3026	12530	1620	Reference	1.600×10^{-10}

The first successful training seed is used for the principal comparison; it is not selected by its error against the reference. The reported final evaluation restores completed checkpoints. Consequently, the training budgets associated with those checkpoints, rather than the defaults of a later evaluation command, identify the computations that produced the weights.

4.4 Analytical plane-flow verification

Before interpreting curved-channel results, I consider plane Poiseuille flow in a straight channel with half-width h. In the baseline units, the analytical solution is

$$u = 1 - (y/h)^2, \qquad v = 0, \qquad p_v = -\frac{2x}{h^2} + C. \tag{4.5}$$

The finite-element relative velocity error is approximately 8.19×10^{-10}. The PINN velocity error is 0.217%, and its gauge-aligned pressure error is 0.180%. The velocity metric includes both components; figure 4.2 displays the streamwise profile for readability. Pressure is evaluated on points distributed along the channel, with each array's mean removed.

The near-exact finite-element result for this polynomial plane flow is consistent with the approximation spaces and simple geometry. It should not be interpreted as the error of the curved-channel reference. The curved problem requires its own mesh-sensitivity assessment.

4.5 Curved-channel carrier-flow results

Figure 4.3 compares the principal neural field with the finite-element field. Both solutions recover the curved high-speed core and the pressure decrease through the channel. Common color limits are used for corresponding speed and pressure panels. The discrepancy panels show that agreement is spatially nonuniform; a small global norm should not be interpreted as uniformly small local error.

The velocity error is 1.510%, and the centered pressure error is 0.863% on the independent field-comparison samples. The maximum absolute velocity discrepancy is approximately

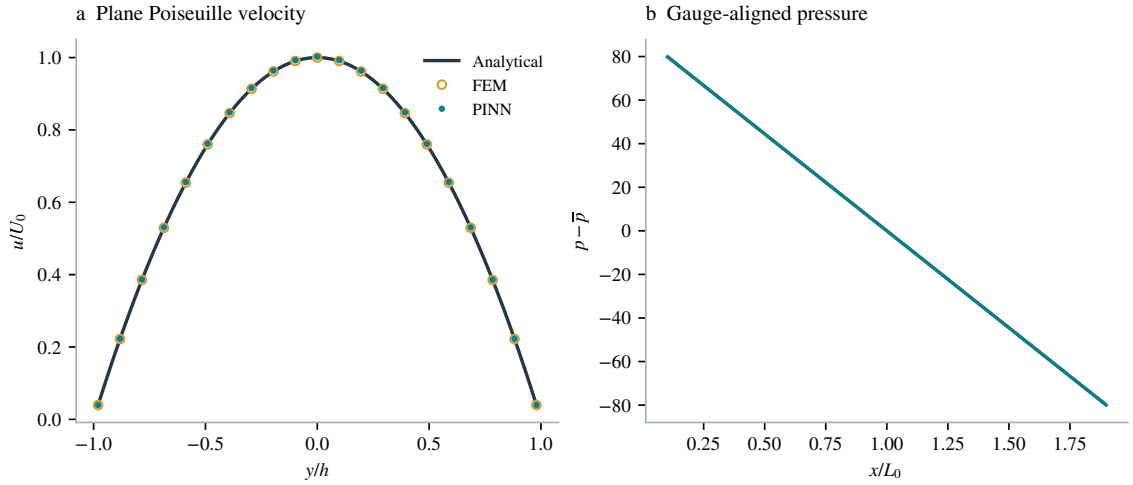


Figure 4.2. Analytical verification of the baseline flow solvers. Panel (a) compares the streamwise velocity profile with the parabolic solution. Panel (b) compares pressure after constant-gauge alignment on spatially distributed samples. Lines and markers retain the stored predictions; no fitted correction is applied.

0.05401. These values support agreement for the specified boundary-value problem. They do not establish zero equation residual: the separately recorded raw continuity and momentum RMS values are approximately 0.1781 and 1.8296 in the baseline equation convention. Those residuals, the field errors, and the flux diagnostic measure different aspects of the approximation.

Table 4.2. Supported baseline numerical comparisons. The relative errors refer to their stated sample sets and normalizations.

Problem	Quantity	Error
Plane Poiseuille PINN	Velocity	0.217%
Plane Poiseuille PINN	Centered pressure	0.180%
Plane Poiseuille FEM	Velocity	8.192×10^{-10}
Curved-channel PINN	Velocity	1.510%
Curved-channel PINN	Centered pressure	0.863%
ForceNet	Normalized drag	0.717%
Section flux	Maximum relative deviation	2.266%
Two finest FEM meshes	Velocity difference	0.173%

4.6 Held-out drag and field comparisons

The normalized ForceNet error is 0.717% on the independent 2,048-sample evaluation used for the reported metric. Figure 4.4 presents velocity and pressure parity together with a separate 1,000-sample drag diagnostic. The plotted drag set and the aggregate 2,048-sample test are

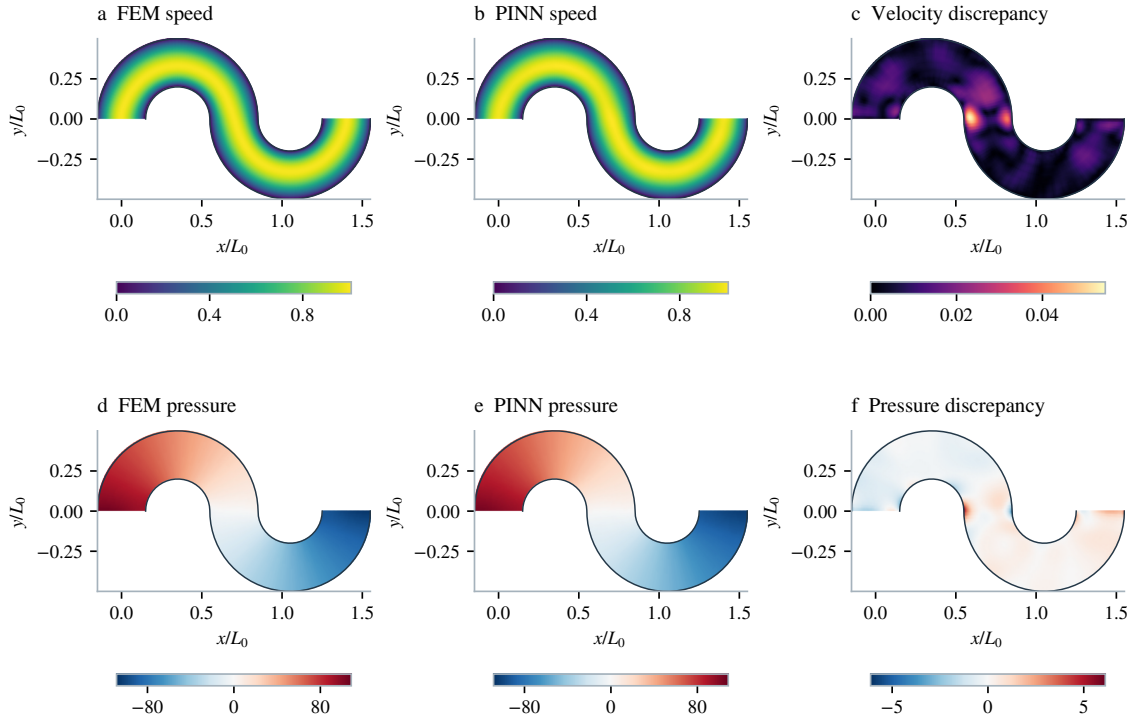


Figure 4.3. Curved-channel carrier field at $\mathrm{Re} = 1$. The speed panels use $|\boldsymbol{u}|/U_0$ and identical scales; the pressure panels use centered viscous-scale pressure and identical scales. Velocity discrepancy is the Euclidean component error, whereas pressure discrepancy is signed. The fields are evaluated on the same stored mapped grid; no spanwise solution is implied.

distinct; they need not have identical numerical errors. This distinction avoids assigning one test's error to another test's points.

The drag comparison supports the network's approximation of the retained creeping-flow law. It cannot establish the accuracy of neglected lift, history, lubrication, or wall corrections [9–11]. Direct analytical Stokes drag also remains an essential baseline for judging the value of a learned approximation of this known relation.

4.7 Seed variation, reference sensitivity, and flux

Two independently trained seeds give velocity errors of 1.510% and 1.429%, and pressure errors of 0.863% and 0.886%, respectively. Their mean velocity error is 1.469%, with a sample standard deviation of 0.057 percentage points. Two seeds are sufficient to reveal some training dependence but are insufficient for a strong statistical uncertainty statement. No confidence interval is inferred.

The maximum section-flux deviation is 2.266%. Figure 4.6 shows the full stored section sequence normalized by the prescribed flux. The variation is finite and should remain visible even though the study's 5% flux criterion is met. Together with boundary RMS of approximately 0.00420, this result provides an integrated check on the velocity field that

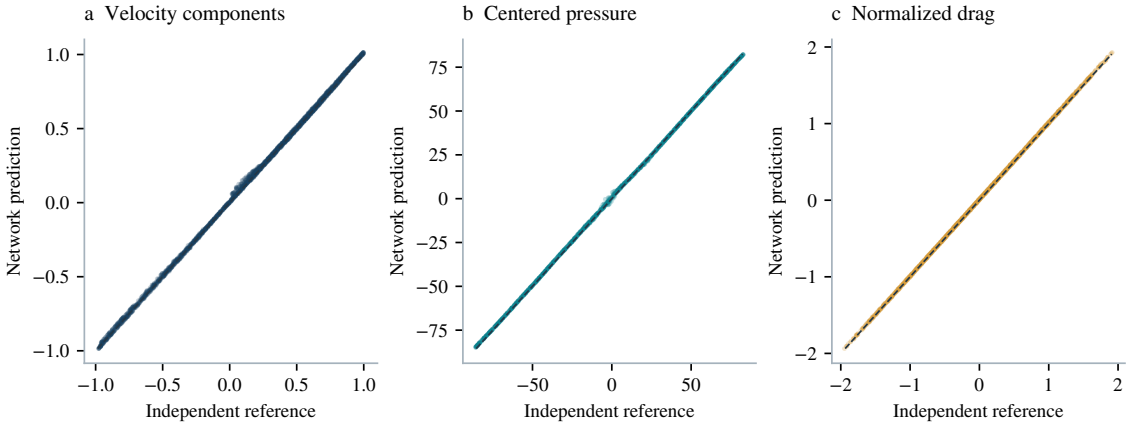


Figure 4.4. Independent numerical comparisons. Velocity components and gauge-aligned pressure are compared with the finite-element reference. The third panel compares normalized learned drag with the specified Stokes-slip response on the stored 1,000-sample diagnostic. The dashed lines indicate equality. The aggregate drag error in table 4.2 comes from a separate 2,048-sample test.

Table 4.3. Field errors for the two completed training seeds.

Seed	Velocity error (%)	Pressure error (%)
42	1.510	0.863
43	1.429	0.886

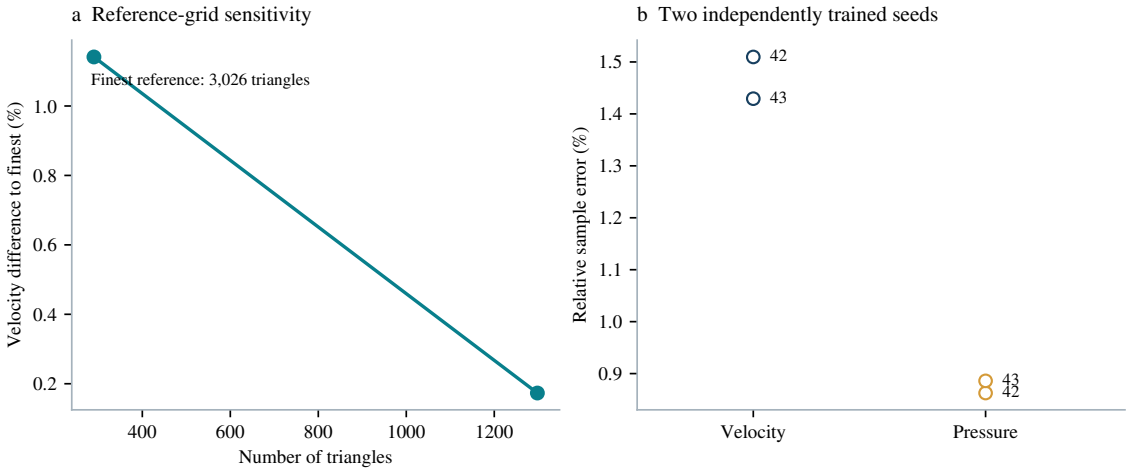


Figure 4.5. Reference and training sensitivity. The left panel compares the two coarser finite-element meshes with the finest reference; the right panel shows the two completed training seeds. The finest mesh is a comparison reference, not an exact solution. Seed labels identify the individual observations rather than confidence bounds.

complements the sample-relative error.

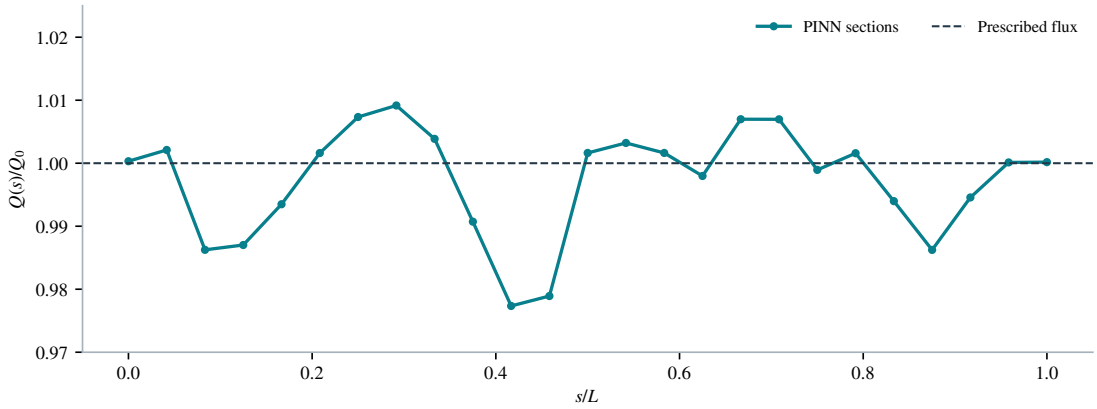


Figure 4.6. Section flux in the curved baseline, normalized by the prescribed value $Q_0 = 4h/3 = 0.2$. The dashed line is the exact constant-flux target, and every stored PINN section is shown. The maximum deviation is 2.266%; the curve has not been rescaled to enforce agreement.

4.8 Particle-transport interpretation

The baseline evaluation completes all 119 particle or tracer integrations: 49 use PINN with ForceNet, 49 use FEM with exact Stokes drag, seven use PINN with exact Stokes drag, and 14 are tracer integrations split equally between the PINN and FEM fields. These counts include the parameter-study cases; the primary endpoint comparison concerns seven initial conditions. The reference velocity interpolator differs from direct finite-element evaluation by 0.177% on its assessment samples. These are useful computational checks, but neither alone proves accurate particle outcomes.

Among the seven baseline particle initial positions, six endpoint classes agree with the reference. The largest width-normalized RMS position discrepancy is approximately 34.97%, and the lowest common-time coverage is approximately 73.55%. Three initial conditions meet all endpoint, position-error and common-time criteria; six meet the endpoint criterion alone. Detailed values are retained in table A.1 so that the scope of the carrier-flow findings is unambiguous. The quantitative conclusions of this chapter concern the carrier field, drag approximation, and the seven tabulated path comparisons.

This outcome is physically and numerically informative. Bulk field agreement and a good approximation of a specified drag law are necessary components, but their combination is not enough to certify wall-sensitive transport. The justified baseline conclusion is that the carrier and drag components agree well with their stated references, while the full trajectory prediction remains an improvement target.

The present comparisons do not isolate a single cause for the trajectory differences. Carrier-field approximation, ForceNet error, reference interpolation, ODE integration, and

event handling are separate candidates. The recorded integrator uses relative and absolute tolerances 2×10^{-6} and 2×10^{-8}, with RK45 or Radau chosen according to the response scale. A factorial comparison holding the integrator and wall-event rule fixed is needed to attribute the discrepancies.

4.9 Chapter conclusions

The baseline establishes a reproducible carrier-flow and drag foundation for the subsequent coupling study. Analytical plane flow, independent curved-channel fields, two completed training seeds, and section flux provide complementary evidence. The results also set a boundary on the claim: the present field accuracy does not establish reliable deposition or all baseline particle endpoints. That boundary becomes more important when reaction forcing and geometric narrowing are introduced.

Chapter 5

Gaussian Parcels in a Straight Channel

5.1 From individual particles to represented populations

Euler–Lagrange methods couple an Eulerian carrier description with Lagrangian particle dynamics [2, 20]. The second stage adds hydrodynamic feedback while retaining a simple carrier geometry. A parcel represents N_i identical spherical particles whose centers are approximated by a common planar state. Its mass and hydrodynamic reaction are weighted by that multiplicity. The Gaussian distribution describes how the represented force interacts with the carrier; it does not resolve the surface of a sphere.

The dimensionless fluid domain is $\Omega = [0,3] \times [0,1]$, periodic in x with stationary no-slip walls at $y = 0$ and $y = 1$. The channel has no open inlet or outlet. The initial fluid velocity is

$$\boldsymbol{u}_0(y) = (6y(1-y), 0), \tag{5.1}$$

whose mean streamwise speed is one. A body-force acceleration $\boldsymbol{f}_d = (12/\mathrm{Re}_H, 0)$ maintains this profile in the unladen steady limit. Parcel initial velocities are zero. The reference depth ratio is $b = 1$, and the evaluated model uses $\mathrm{Re}_H = 20$, $R = 1000$, six parcels, and Gaussian width $\sigma = 0.14$.

5.2 Coupled equations

The fluid equations use the inertial pressure convention defined in equation (3.8):

$$\nabla \cdot \boldsymbol{u} = 0, \tag{5.2}$$

$$\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p_i + \frac{1}{\mathrm{Re}_H} \nabla^2 \boldsymbol{u} + \boldsymbol{S} + \boldsymbol{f}_d. \tag{5.3}$$

The parcel equations are

$$\dot{\boldsymbol{q}}_i = \boldsymbol{v}_i, \qquad \dot{\boldsymbol{v}}_i = \frac{\boldsymbol{U}_i - \boldsymbol{v}_i}{\mathrm{St}_i} + \boldsymbol{g}_{\mathrm{eff}}, \qquad \boldsymbol{F}_i = \frac{m_i}{\mathrm{St}_i}(\boldsymbol{U}_i - \boldsymbol{v}_i). \tag{5.4}$$

Here $\boldsymbol{g}_{\text{eff}}$ is a specified effective acceleration, including any buoyancy adjustment in the physical mapping. The reaction source contains the hydrodynamic drag only. Depositing parcel gravity again as fluid drag would count a different force twice.

5.3 Normalized Gaussian transfer

Let $\boldsymbol{x}_a$ denote the fixed transfer-quadrature locations and ΔA_a their area weights. A periodized raw Gaussian is evaluated between $\boldsymbol{x}_a$ and each parcel center. In the two-dimensional channel, its structure is

$$G_{ai} = \sum_{k\in\mathbb{Z}} \exp\left[-\frac{(x_a - q_{ix} + kL_x)^2 + (y_a - q_{iy})^2}{2\sigma^2}\right]. \tag{5.5}$$

The implementation retains a finite image set with distant tails beyond eight widths after choosing the nearest image. A global Gaussian prefactor is unnecessary because normalization removes it. Define

$$Z_i = \sum_a G_{ai}\Delta A_a, \qquad W_{ai} = G_{ai}/Z_i, \qquad \sum_a W_{ai}\Delta A_a = 1. \tag{5.6}$$

Both fluid interpolation and force spreading use these weights:

$$\boldsymbol{U}_i = \sum_a W_{ai}\boldsymbol{u}_a\Delta A_a, \qquad \boldsymbol{S}_a = -\sum_i W_{ai}\boldsymbol{F}_i. \tag{5.7}$$

Near a wall, Z_i accounts for the actual bounded-domain quadrature. At an arbitrary collocation point, the numerator uses that point, while the denominator remains the value from the fixed transfer quadrature.

The evaluated channel uses tensor-product midpoint quadrature. Its points are $x_a = (a+1/2)3/16$ and $y_b = (b+1/2)/16$ for $a,b = 0,\ldots,15$. Thus $\Delta x = 0.1875$, $\Delta y = 0.0625$, and every point has area weight $\Delta A = 3/256$. Exact normalization on this grid is a property of the discrete operator; spatial resolution is assessed separately.

5.3.1 Force and work identities

Substituting equation (5.7) into the quadrature sum gives

$$\sum_a \boldsymbol{S}_a\Delta A_a = -\sum_i \boldsymbol{F}_i, \tag{5.8}$$

and

$$\sum_a \boldsymbol{u}_a\cdot\boldsymbol{S}_a\Delta A_a = -\sum_i \boldsymbol{U}_i\cdot\boldsymbol{F}_i. \tag{5.9}$$

The combined interphase contribution to fluid and parcel kinetic-energy rates is therefore

$$\mathscr{P}_{\mathrm{drag}} = \sum_i (\boldsymbol{v}_i - \boldsymbol{U}_i) \cdot \boldsymbol{F}_i = -\sum_i \frac{m_i}{\mathrm{St}_i} |\boldsymbol{U}_i - \boldsymbol{v}_i|^2 \leq 0. \tag{5.10}$$

These are semidiscrete properties of the shared transfer. Paired Eulerian–Lagrangian transfer operators provide related methodological background [15]; the identities above follow specifically from equations (5.6) and (5.7). A trained PINN with nonzero residual need not satisfy the complete global momentum or energy balance exactly. External driving can also make total kinetic energy increase even though the drag contribution is dissipative.

5.4 Neural representation and hard constraints

Two MLPs, each with three hidden layers of 32 tanh units, represent the fluid correction and parcel histories. The fluid network uses wall-normal position, time, and three periodic Fourier harmonics in x. The ansatz is

$$\boldsymbol{u}(\boldsymbol{x},t) = \boldsymbol{u}_0(y) + \frac{t}{T} y(1-y) \boldsymbol{n}_u(\boldsymbol{x},t). \tag{5.11}$$

It enforces the initial velocity and stationary wall velocity, following the principle of building constraints into the approximation [26, 27]. Periodicity follows from the Fourier features. A time-dependent pressure offset is removed at $(0,0)$; no independent initial-pressure condition is imposed.

The parcel network uses the relaxation time in its output envelopes:

$$b_i(t) = 1 - \exp(-t/\mathrm{St}_i), \qquad a_i(t) = t - \mathrm{St}_i b_i(t), \tag{5.12}$$

$$\boldsymbol{q}_i(t) = \boldsymbol{q}_{i0} + t\boldsymbol{v}_{i0} + a_i(t)\boldsymbol{n}_{q,i}(t), \qquad \boldsymbol{v}_i(t) = \boldsymbol{v}_{i0} + b_i(t)\boldsymbol{n}_{v,i}(t). \tag{5.13}$$

The envelope imposes the initial states and represents the known response scale. It supplies no target fluid velocity or target coupled trajectory. Position and velocity remain separate trainable functions, linked by $\dot{\boldsymbol{q}}_i - \boldsymbol{v}_i$ in the objective.

The weighted mean-square residual objective assigns weights 1 to fluid momentum, 2 to continuity, 5 to parcel kinematics, 1 to parcel acceleration, and 100 to center confinement. All calculations in this formulation use float64. Figure 5.1 shows the two neural representations, their shared transfer, and the common training objective. The default calculations are collision-free; independently tested impact primitives are not treated as a validated collisional suspension model.

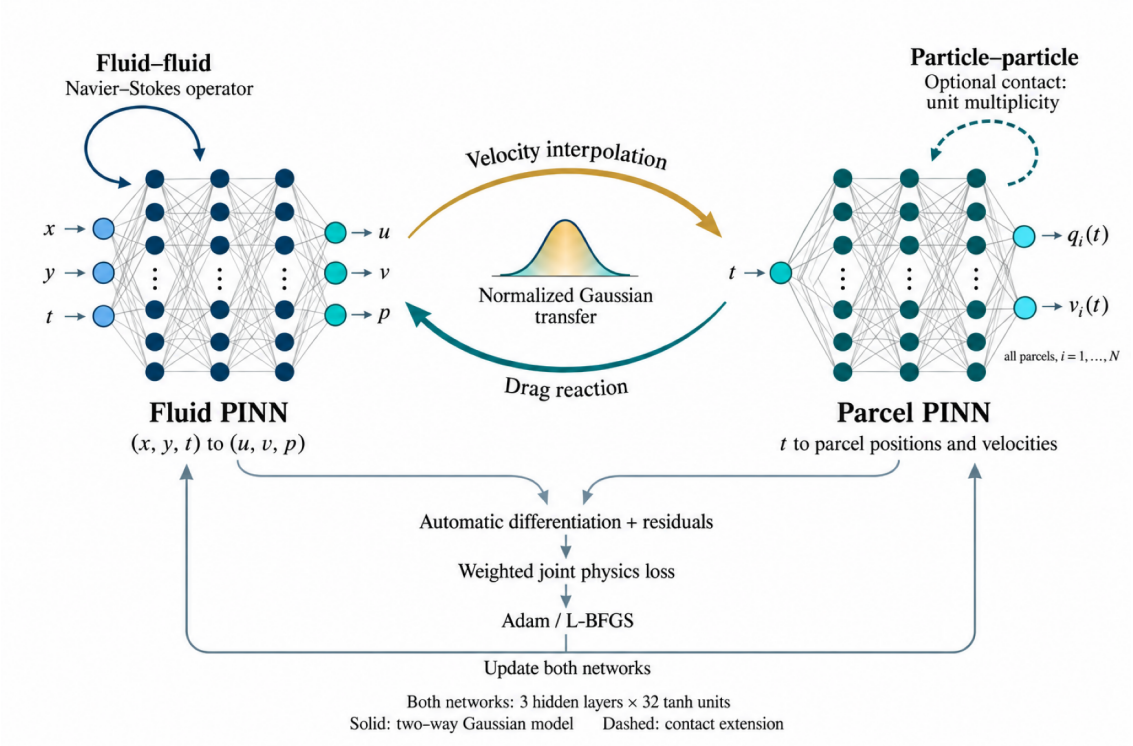


Figure 5.1. Coupled PINN architecture and interaction pathways. The fluid and parcel networks exchange sampled velocity and hydrodynamic reaction through the same normalized Gaussian. Both feed a common residual objective and receive joint parameter updates. The outer loops denote interaction operators, not recurrent neural layers. Solid paths describe the evaluated two-way Gaussian model; the dashed particle–particle branch is the optional unit-multiplicity contact extension. Neuron symbols are representative, and the parcel outputs collect all planar positions and velocities.

5.4.1 Interaction pathways and joint training

Fluid–fluid interaction enters through the Navier–Stokes differential operator, while fluid–parcel and parcel–fluid exchange use the paired Gaussian transfer in equation (5.7). Adding direct particle–particle interaction to two-way exchange gives the four-way extension of this hierarchy [1, 14]. The optional `soft_sphere` branch evaluates pair and wall contact for physical spheres with $N_i = 1$. The reported Gaussian-parcel cases use `collision=none`; their contact acceleration is zero. This distinction is shown by the dashed branch in figure 5.1.

Algorithm 5.1 gives the computational sequence. Let $\boldsymbol{r}_m$, r_d, $\boldsymbol{r}_q$, $\boldsymbol{r}_v$, and r_c denote momentum, divergence, kinematic, acceleration, and confinement residuals. Let $\mathscr{M}(r)$ denote the mean squared value over every sample and component of a residual array. The implemented objective is

$$\mathscr{L}(\theta,\phi) = \mathscr{M}(\boldsymbol{r}_m) + 2\mathscr{M}(r_d) + 5\mathscr{M}(\boldsymbol{r}_q) + \mathscr{M}(\boldsymbol{r}_v) + 100\mathscr{M}(r_c). \tag{5.14}$$

Automatic differentiation propagates through both networks, the position-dependent Gaussian weights, interpolation, and reaction spreading.

Algorithm 5.1 Joint PINN training with an optional contact branch

1. Select the interaction mode and initialize θ, ϕ, and $(\boldsymbol{x}_a, \Delta A_a)$. For physical-sphere contact, set $N_i = 1$ when computing m_i. Apply the Fourier features and constrained ansatzes in equations (5.11) and (5.13).
2. At sampled times, evaluate $(\boldsymbol{u}, p)$ and all $(\boldsymbol{q}_i, \boldsymbol{v}_i)$. Compute G_{ai}, Z_i, and W_{ai} on the fixed quadrature using equation (5.6).
3. Interpolate $\boldsymbol{U}_i$ and compute $\boldsymbol{F}_i = m_i(\boldsymbol{U}_i - \boldsymbol{v}_i)/\mathrm{St}_i$. Spread its negative reaction to fluid evaluation points with the same normalizers Z_i.
4. Set $\boldsymbol{a}_i^c = 0$ for the Gaussian-parcel model. If physical-sphere contact is selected, compute $\boldsymbol{a}_i^c$ from the clipped spring–dashpot pair and wall forces divided by the physical sphere mass.
5. Differentiate the fields and histories to form the fluid, kinematic, confinement, and acceleration residuals, with
$$\boldsymbol{r}_{v,i} = \dot{\boldsymbol{v}}_i - \frac{\boldsymbol{U}_i - \boldsymbol{v}_i}{\mathrm{St}_i} - \boldsymbol{g}_{\mathrm{eff}} - \boldsymbol{a}_i^c.$$
The fluid reaction contains hydrodynamic drag only.
6. Assemble equation (5.14) and update (θ, ϕ) jointly. Use fresh batches during Adam and a fixed batch during L-BFGS; select the saved checkpoint using the fixed audit objective. Evaluate held-out residuals separately from training.

5.5 Reduced Stokes-response verification

The first reduced test, V01, solves the constant-forcing relaxation problem with dimensionless response time one:

$$\dot{v} = 1 - v, \qquad \dot{z} = v, \qquad v(0) = z(0) = 0. \tag{5.15}$$

Its analytical response is

$$v = 1 - e^{-t}, \qquad z = t - 1 + e^{-t}. \tag{5.16}$$

The reference arrays are excluded from training and used only after the physics-constrained fit. The relative velocity and displacement errors are 0.066% and 0.024%, with independent ODE residual RMS approximately 1.35×10^{-3}. Both the stated relative-error and residual criteria are satisfied. Figure 5.2 shows the stored predictions against the analytical response.

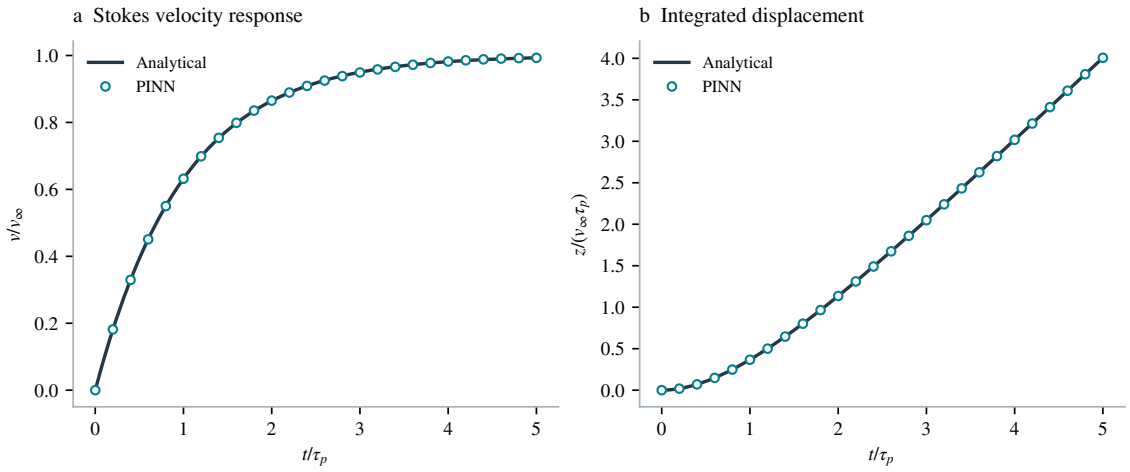


Figure 5.2. V01 reduced Stokes response. The PINN is fitted to the ODE residual and initial conditions; analytical values are held out. Open markers are subsampled stored predictions for legibility; the metrics use every evaluation sample. This test verifies the reduced response equation, not the full two-way channel.

5.6 Two-way momentum-exchange verification

The second reduced problem, V02, considers spatially uniform fluid and particle velocities, fluid mass one, parcel mass loading $\beta = 0.25$, and response time $\mathrm{St} = 0.2$. With $u(0) = 0$ and $v(0) = 1$,

$$\dot{u} = -\frac{\beta}{\mathrm{St}}(u - v), \qquad \dot{v} = \frac{u - v}{\mathrm{St}}. \tag{5.17}$$

The solution has constant total momentum $u + \beta v = 0.25$ and decaying slip. The relative fluid- and particle-velocity errors are 0.096% and 0.238%, respectively. The independent residual RMS is approximately 6.12×10^{-3}.

In this reduced PINN, the output relation $v = 1 - 4u$ imposes total momentum exactly. Its

zero momentum error is consequently a hard-constraint check, not an independent demonstration that a general coupled solution conserves momentum. The nontrivial checks are the predicted relaxation and its ODE residual. Figure 5.3 also shows the kinetic-energy decay, consistent with the sign in equation (5.10).

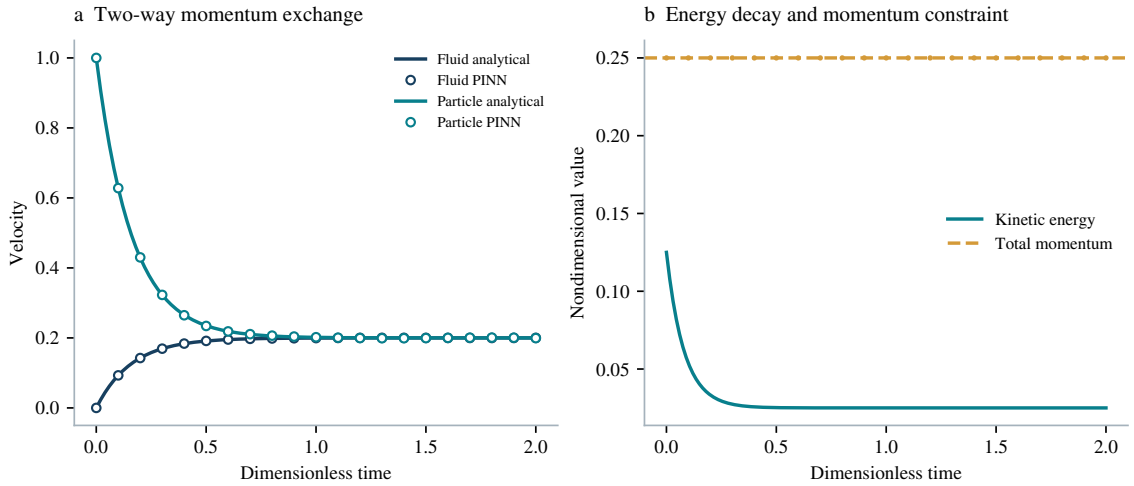


Figure 5.3. V02 uniform two-way relaxation. The two velocities approach a common value while kinetic energy decreases. The total-momentum line is exact by the imposed output relation. Velocity agreement and the independently sampled ODE residual assess whether the learned temporal response satisfies the governing problem.

5.7 Kernel and carrier-flow verification

5.7.1 Normalization, overlap, and parcel splitting

The bounded-domain normalization test V03 probes periodic-seam and wall locations. Its maximum recorded discrepancy is approximately 8.44×10^{-15} across the checks included in that test. The displayed integrals in figure 5.4 differ from one by only a few units of machine precision. The larger summary discrepancy also includes the constant-field sampling check and should not be assigned to an individual integral in the figure.

V04 compares a three-dimensional product quadrature with an independent free-space Gaussian overlap array. The relative difference is approximately 1.29×10^{-15}. The CSV column retained from the numerical workflow uses a generic prediction name; here it is correctly identified as a quadrature result rather than a learned PINN field. This is a kernel-integral test and does not establish a three-dimensional channel solution.

V09 splits a fixed population of 10,000 particles into 10, 50, and 100 coincident parcels. The maximum force-density difference is approximately 2.66×10^{-15}. Because the split centers coincide and total mass is unchanged, this tests deterministic representation invariance. It is not a convergence study of independently dispersed parcel populations.

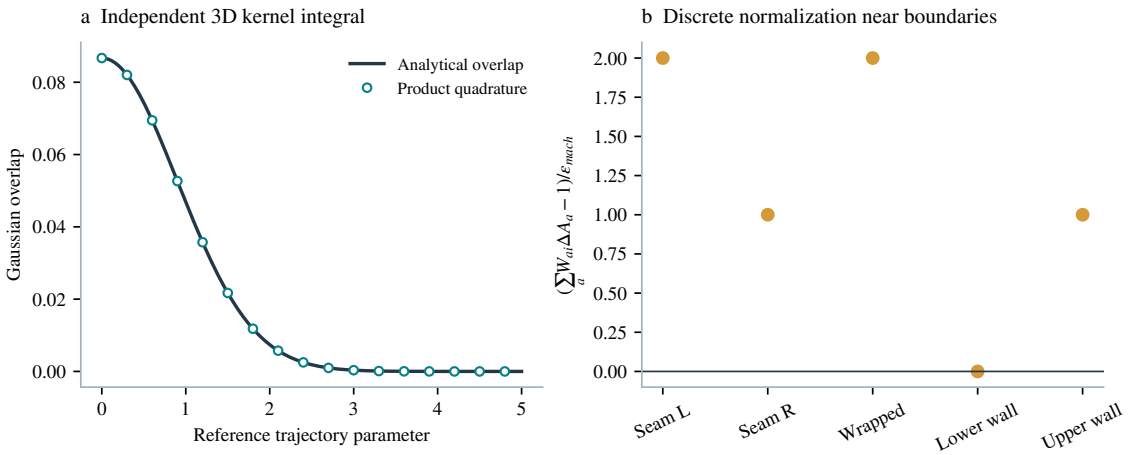


Figure 5.4. Gaussian transfer checks. Panel (a) shows V04's independent three-dimensional overlap integral and product quadrature. Panel (b) shows the stored V03 normalization values divided by machine precision. The kernels are verified on their declared quadrature; neither panel is a flow-field prediction.

5.7.2 Unsteady incompressible flow

The Taylor–Green vortex family supplies historical background [36], while velocity–pressure PINN flow benchmarks provide related numerical context [34]. V10 uses the following explicit two-dimensional periodic form and pressure gauge, with viscosity $\nu = 0.01$ over $0 \le t \le 5$. The exact velocity is

$$u = \sin x \cos y\, e^{-2\nu t}, \qquad v = -\cos x \sin y\, e^{-2\nu t}, \tag{5.18}$$

and the gauge-aligned pressure is

$$p = \frac{1}{4}(\cos 2x + \cos 2y - 2)e^{-4\nu t}. \tag{5.19}$$

The network imposes the initial velocity and initial gauge-aligned pressure through

$$u_\theta = \sin x \cos y + (t/5)n_u, \tag{5.20}$$

$$v_\theta = -\cos x \sin y + (t/5)n_v, \tag{5.21}$$

$$p_\theta = \tfrac{1}{4}(\cos 2x + \cos 2y - 2) + (t/5)[n_p(x,y,t) - n_p(0,0,t)]. \tag{5.22}$$

The trainable corrections n_u, n_v, n_p use periodic features; the analytical temporal decay is held out of training. This evaluates the learned response within the stated constrained representation. Independent sample-relative errors are 0.218% for u, 0.187% for v, and 0.215% for pressure. The independent residual RMS is approximately 7.38×10^{-4}. These results support the unsteady carrier implementation for this smooth periodic test.

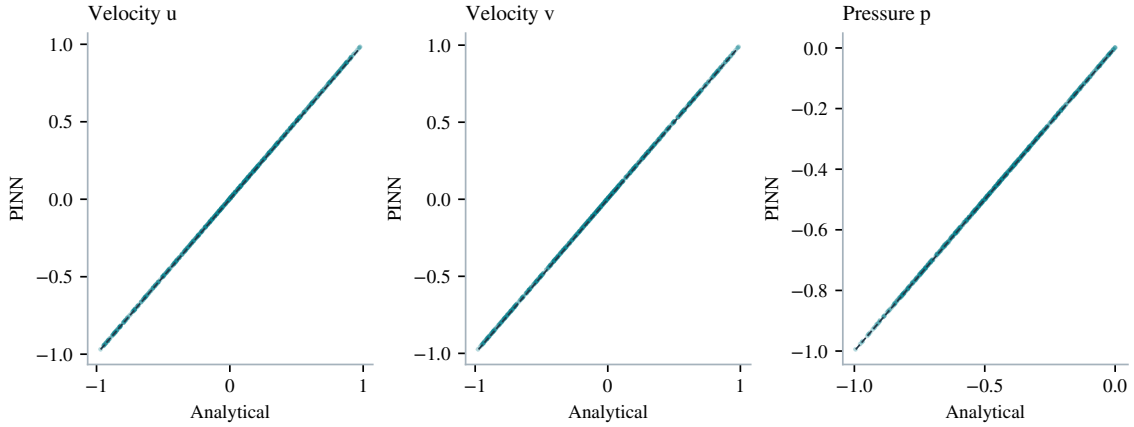


Figure 5.5. V10 Taylor–Green verification on independent space–time samples. Each panel compares the predicted quantity with its analytical value, using the same pressure gauge. The test contains no parcels and establishes carrier-flow verification rather than coupled particle-transport validation.

Table 5.1. Accepted solution-observable errors in the reduced Gaussian-model verification problems. Residual criteria are checked separately.

Test	Observable	Relative error
V01	Stokes velocity	0.066%
V01	Integrated displacement	0.024%
V02	Fluid velocity	0.096%
V02	Particle velocity	0.238%
V10	Taylor–Green u	0.218%
V10	Taylor–Green v	0.187%
V10	Taylor–Green p	0.215%

5.8 Coupled computations and accuracy assessment

The evaluated coupled channel uses one training seed and four combinations of Stokes number and mass loading. The training schedule is 1,200 Adam steps followed by a 200-iteration L-BFGS budget; the transfer quadrature is 16×16. All four calculations complete with finite trained checkpoints and a recorded budget-exhaustion stopping status. The coupled implementation produces computed carrier fields and parcel trajectories. Its numerical accuracy is assessed independently using the measured momentum and parcel-acceleration RMS values in table 5.2. The momentum RMS ranges from 0.0407 to 0.3854, compared with the declared reference criterion of 0.03. These residuals describe the accuracy of the saved fields; optimizer convergence requires its own assessment.

Table 5.2. Coupled straight-channel residuals. The reference criterion for each required held-out componentwise RMS is 0.03. The final column is momentum RMS divided by this criterion; a ratio greater than one indicates a residual above the reference value.

St	β	Momentum RMS	Parcel RMS	Momentum/0.03
0.08	0.05	0.3854	0.0476	12.85
0.20	0.05	0.1372	0.0267	4.57
0.20	0.15	0.3526	0.0834	11.75
0.60	0.05	0.0407	0.0017	1.36

The numerical assessment also requires all centers to remain inside the channel, a trained finite checkpoint without training error, local volume fraction below 0.01, and slip Reynolds number no greater than one. The observed maximum slip Reynolds numbers range from approximately 0.234 to 0.642. These satisfy that loose screening bound but do not establish the asymptotically small-slip regime required for accurate Stokes drag. The reduced tests and coupled-channel results address different levels of the calculation. The coupled figures document computed fields and parcel motion; quantitative loading, sedimentation and deposition trends require a matched accuracy assessment. The physical model also samples the disturbed fluid without an undisturbed-velocity correction [17, 18], so successful future residual screening would still require independent assessment of the drag closure.

All ten component tests in the evaluated record pass their individual criteria. Tests V05–V08 concern impact algebra and a frozen-rate sampling primitive, which are outside the collision-free channel study and are not used to validate its dynamics. The literature catalogue likewise supplies no matched physical-validation result for the Gaussian channel. These scope limits distinguish component verification from coupled-field accuracy.

The held-out fluid assessment uses nine times and 79 independently drawn spatial points per time, giving 711 space–time locations. Four times explicitly probe $t/T = 0$, 0.01, 0.05 and 1; the remaining times and spatial points follow the independent seed 991. Parcel residuals

use the same nine times and all six parcels. For momentum, RMS is the square root of the mean of squared residual components over times, locations and the two velocity components. Parcel RMS analogously averages over times, parcels and components. The 0.03 limit applies to these dimensionless componentwise RMS values, not to a percentage field error or a mean vector norm.

5.9 Computed parcel motion and carrier response

The following figures document the finite computed predictions with the accuracy scope quantified in table 5.2. I use the $\mathrm{St} = 0.6$, $\beta = 0.05$ case to document the computed motion. Among the four evaluated cases it has the smallest held-out momentum residual and the smallest parcel-acceleration residual. This selection identifies the more internally consistent available calculation; it is not evidence of a general advantage of larger Stokes number. The complete four-case comparison remains in table 5.2.

Table 5.3. Actual settings of the displayed six-parcel computation. Masses use the planar convention of equation (3.9); quantities are dimensionless unless stated otherwise.

Quantity	Value
Re_H, R, b	20, 1000, 1
β, parcel count, seed	0.05, 6, 7
σ, T	0.14, 1
$\boldsymbol{g}_{\mathrm{eff}}$	$(0, -0.1)$
St, d/H	0.60, 0.023238
Multiplicity N_i	3.804966
Represented mass m_i	0.025000
Single-sphere mass in the planar scale	0.00657036
Initial parcel velocity	$(0,0)$

Table 5.4. Initial centers of the six displayed parcels, read from the stored states. All initial velocities are zero.

Parcel	x_0/H	y_0/H
P1	0.887605	0.698607
P2	1.053254	0.362604
P3	0.530183	0.686777
P4	0.205792	0.660614
P5	1.076776	0.483967
P6	0.533336	0.389213

For this case the kinematic and acceleration RMS values are approximately 6.81×10^{-4}

and 1.65×10^{-3}, respectively. The carrier momentum RMS remains 0.0407, above 0.03. The following figures therefore illustrate the model's computed response without upgrading it to a validated coupled solution or a quantitative loading/deposition trend.

5.9.1 Transport and parcel acceleration

Figure 5.6 shows four stored times over $0 \leq t \leq 1$. The same six parcels are identified by consistent colors. They move downstream while remaining inside the channel; their speed starts from zero and approaches the local sampled carrier speed over the response time. The carrier remains predominantly streamwise, with a high-speed core and stationary-wall constraints.

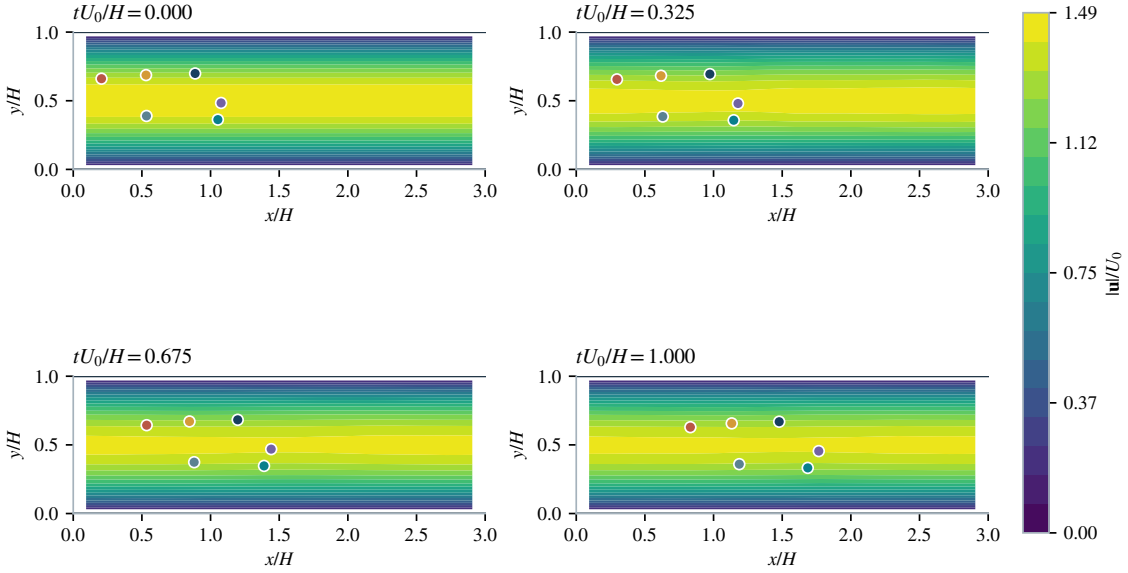


Figure 5.6. Computed transport for St = 0.6, $\beta = 0.05$, seed 7. Background colors show carrier speed at four stored times; matching marker colors identify the six parcels. Markers are illustrative centers, not physical sphere sizes. Colors interpolate the stored 16×16 carrier samples using one common scale; the wall lines identify the prescribed no-slip boundaries. This computed response has momentum RMS 0.0407; the accuracy criterion and all four case outcomes are given in table 5.2.

The trajectories and component velocities in figure 5.7 resolve different aspects of this motion. Streamwise parcel velocities increase from zero to approximately 0.936–1.076 at $t = 1$. All six centers move down by approximately 0.029–0.032 channel heights under the retained effective acceleration and sampled carrier motion. None contacts a wall during this interval. Thus these paths provide no deposition efficiency or wall-capture result.

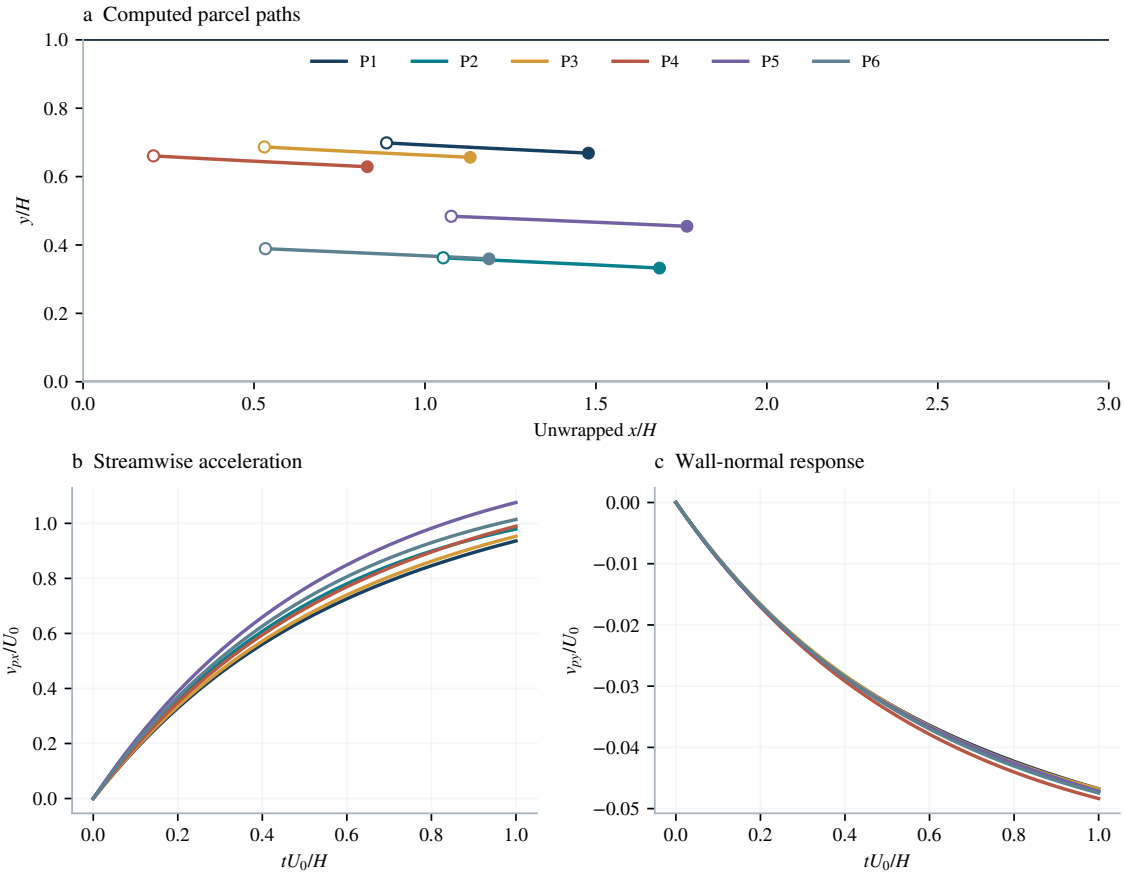


Figure 5.7. Stored parcel paths and velocity histories for the same $\mathrm{St} = 0.6$, $\beta = 0.05$ case. Open markers indicate initial centers and filled markers final centers. Streamwise positions are unwrapped. The lower panels retain the independently predicted velocity outputs, whose consistency with position is assessed by the reported kinematic residual. No trajectory is clipped, replaced by an analytical curve, or removed from the six-parcel set.

5.9.2 Evolution of the represented Gaussian clouds

The effective local fraction is

$$\alpha(\boldsymbol{x},t) = \sum_i \frac{m_i}{R} W_i(\boldsymbol{x},t). \tag{5.23}$$

Figure 5.8 evaluates this expression using the stored parcel centers and the original fixed-quadrature normalizers. The field shows how the represented populations translate and overlap as their centers move. The width is fixed at $\sigma = 0.14$; the display does not imply physical diffusion, collisions, or growth of the individual clouds.

The total fraction integrated on the transfer grid is $\sum_a \alpha_a \Delta A_a = 1.5 \times 10^{-4}$ at every stored time, consistent with total represented mass 0.15 and $R = 1000$. Reconstructing the kernel from the saved states reproduces both the stored fraction and the sampled streamwise fluid velocity to floating-point precision. This checks transfer and postprocessing consistency, rather than independently validating the coupled flow equations.

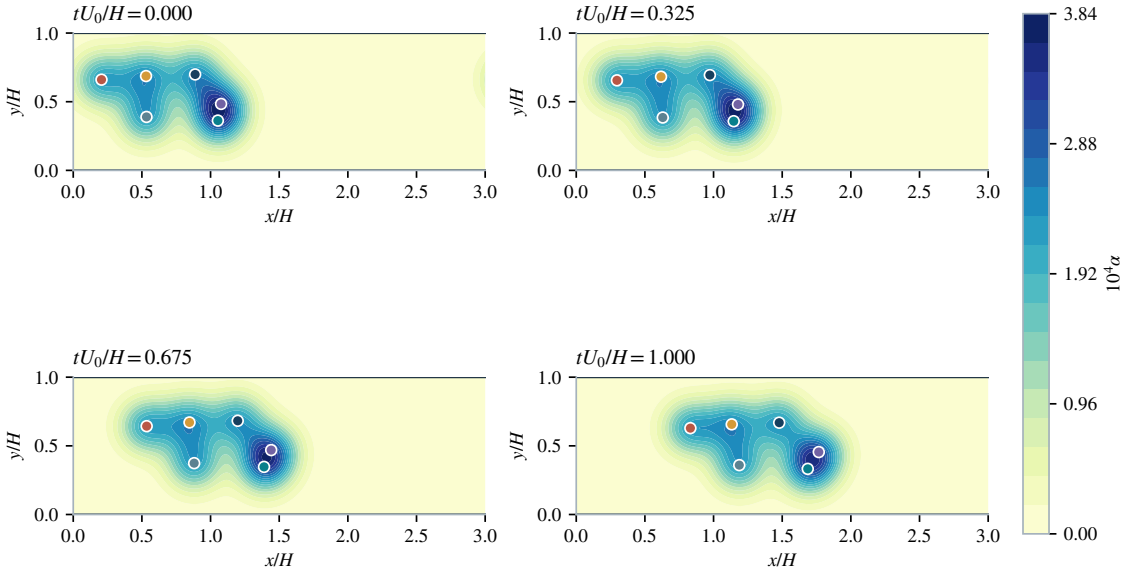


Figure 5.8. Evolution of the effective Gaussian fraction for the selected parcel run. The displayed quantity is $10^4\alpha$ and uses a common color scale. The smooth display evaluates the specified kernel at additional plotting points using stored centers and the original transfer-grid normalization; it is not a refined fluid solution. All six clouds are included. The field represents population weights, not resolved particle surfaces or measured concentration.

5.9.3 Carrier response

At the final time the computed streamwise correction relative to the imposed initial profile is negative at every stored carrier sample, ranging from approximately -0.0701 to -0.00324. The wall-normal carrier component is much smaller than the streamwise speed. Figure 5.9

displays both components while retaining the actual parcel positions. The streamwise correction is a change relative to an initial condition, not an error against an independent two-way reference. Its spatial pattern can be inspected, but the unresolved momentum residual prevents assigning it a certified physical accuracy.

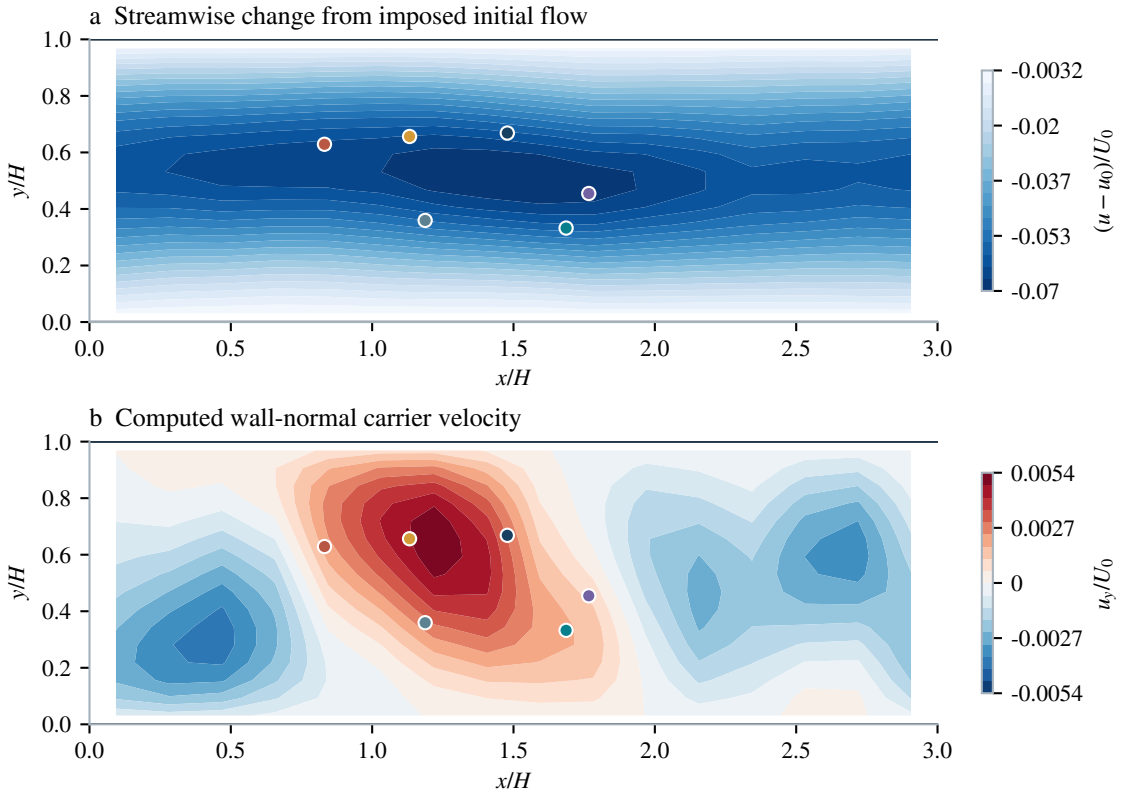


Figure 5.9. Computed carrier response at $tU_0/H = 1$ for the selected run. Panel (a) shows the change of streamwise velocity from the imposed initial profile $u_0 = 6y(1-y)$; panel (b) shows the predicted wall-normal fluid velocity. Each component has its own labeled color scale. The contours interpolate stored transfer-grid samples; they neither represent a resolved particle wake nor an independent error map.

5.9.4 Energy and interphase exchange

Figure 5.10 shows total kinetic energy, drag dissipation, and both recorded balance discrepancies. The computed energy first falls from approximately 1.8000 to 1.7172 and then rises to 1.7329. The late increase is compatible with the presence of imposed driving; a driven channel does not require monotonically decreasing total energy. The drag contribution remains dissipative: D_{drag} decreases from approximately 0.3993 to 0.01338 and stays nonnegative.

The discrepancy panels are retained alongside these observations. In particular, the energy-balance discrepancy changes sign and the momentum quantity is a nonnegative vector norm. Displaying these definitions separately avoids labeling a signed curve as an absolute error. The finite discrepancies and the independent residual screen continue to

limit quantitative interpretation. The plotted energy and dissipation are internally consistent numerical diagnostics, not a claim of exact global conservation by the trained solution.

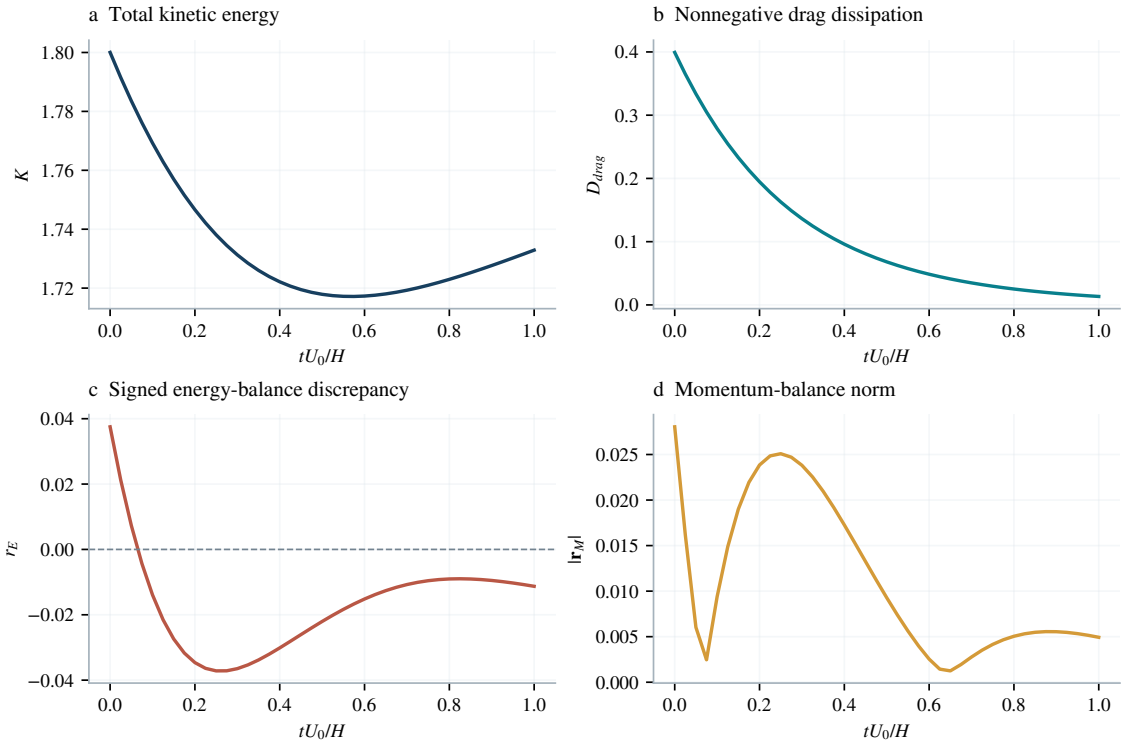


Figure 5.10. Recorded diagnostics for the selected coupled parcel run. Total kinetic energy includes carrier and represented parcel contributions. The plotted drag dissipation is $D_{\mathrm{drag}} \geq 0$, so its contribution to interphase power is $-D_{\mathrm{drag}}$. The energy discrepancy is signed, whereas the momentum discrepancy is a vector norm. No normalization, offset removal, or smoothing is applied to the balance curves.

5.10 Chapter conclusions

The Gaussian formulation establishes exact transfer-quadrature identities and successful reduced tests for relaxation, kernel normalization, and unsteady carrier flow. The strongest supported results are these independently evaluated components. Further optimization, transfer-grid and time-sampling refinement, multiple seeds, and a matched independent coupled reference are required before interpreting the fully coupled channel quantitatively. The next chapter therefore formulates a controlled constricted-channel extension rather than extending unsupported transport claims.

Chapter 6

Proposed Constricted-Channel Study

6.1 Purpose and status of the extension

A constriction introduces a varying gap, a changing cross-sectional mean velocity at fixed flux, and stronger sensitivity to wall treatment. It provides a natural next problem after the constant-width baseline and the Gaussian coupling study. It also introduces new requirements that cannot be met by merely changing the appearance of a straight-channel figure.

This chapter defines the geometry, governing problem, and evaluation design for that extension. The geometry drawings are mathematical illustrations. No constricted-channel velocity, pressure, particle trajectory, or deposition rate has been computed for the accepted result set. The proposed parameter values below are study-design choices rather than measured or completed results.

The obstruction is static and represented by the walls. A progressively growing particle deposit, moving obstacle, or porous plug would require additional evolution laws and is outside this design. I use the term *constriction* for the prescribed narrowing to keep this geometric effect separate from dynamic clogging.

6.2 Parameterized geometry

Let the nominal channel height be H and let the walls be $y = \pm h(x)$. A compact, smooth narrowing centered at x_c is defined by

$$h(x) = \frac{H}{2}[1 - \delta f(\xi)], \qquad \xi = \frac{x - x_c}{\ell}, \qquad 0 \le \delta < 1, \tag{6.1}$$

where

$$f(\xi) = \begin{cases} \left[\dfrac{1+\cos(\pi\xi)}{2}\right]^2, & |\xi| < 1, \\ 0, & |\xi| \ge 1. \end{cases} \tag{6.2}$$

The profile and its first three derivatives match the constant-height segments at the joins. Its minimum gap is $H_t = (1-\delta)H$. The initial design uses $L_x/H = 8$, $x_c/H = 4$, and $\ell/H = 1.2$, with nominal severities $\delta \in \{0, 0.2, 0.4\}$. These values define a controlled sequence from the unblocked channel to moderate narrowing. The actual reference study must check that upstream and downstream lengths are sufficient.

Figure 6.1 displays the geometry and height profiles. Only geometric quantities are plotted. In particular, the height curves must not be interpreted as velocity or pressure results.

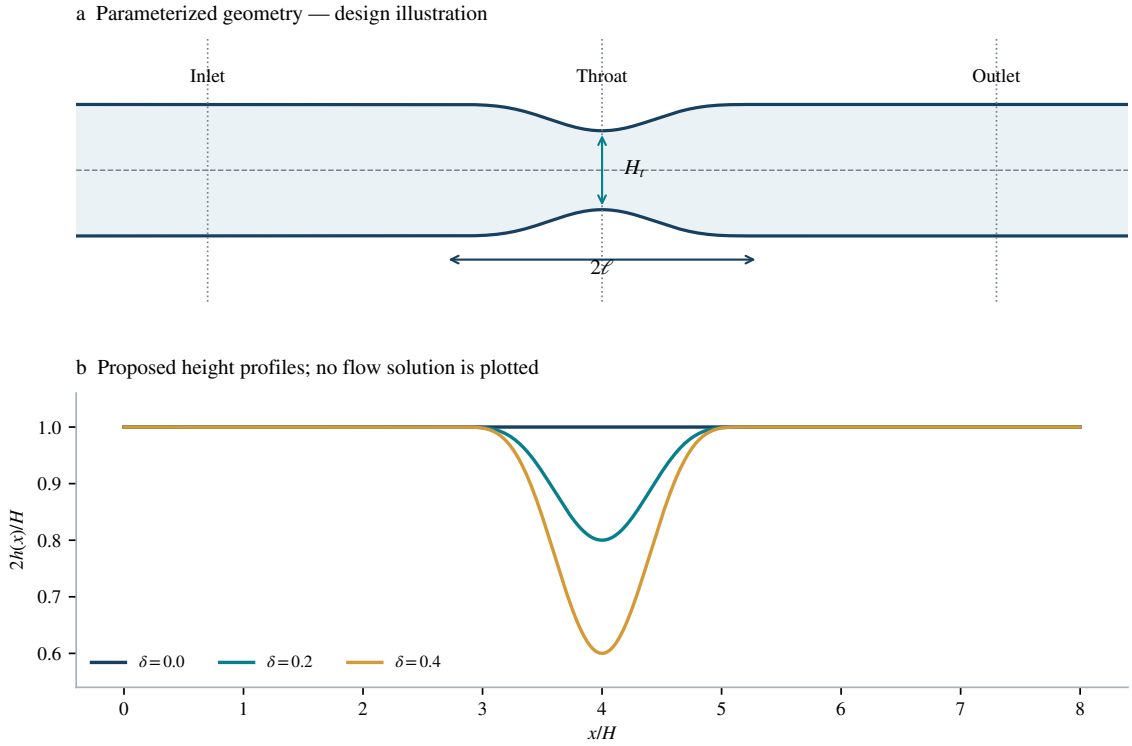


Figure 6.1. Proposed constriction. Panel (a) defines the throat, characteristic length, and reference sections. Panel (b) shows the mathematical height profiles for the proposed severity values. The illustrated channel has $L_x/H = 8$ and $\ell/H = 1.2$; no fluid or particle solution is represented.

6.3 Carrier-flow boundary conditions

For an open-channel extension, the governing equations retain the inertial pressure scale and the same Newtonian constitutive law. I prescribe a parabolic inlet velocity with a specified flux Q and no slip on both walls. A downstream traction condition provides an outlet reference rather than imposing a second velocity profile. The finite-element reference and the PINN must implement exactly the same outlet condition and pressure convention.

This is a different boundary-value problem from both earlier stages. The curved baseline has prescribed velocities at both ends, whereas the straight Gaussian channel is periodic and body-force driven. Comparisons across these stages must therefore identify which changes

follow from geometry and which follow from the drive or boundary conditions. A useful control is a straight open channel with the same inlet and outlet treatment as the constricted case.

For unsteady particle introduction, the initial carrier field should be an independently converged unladen solution in the constricted domain. Imposing the original straight-channel profile throughout the narrowed region would violate the new wall geometry. An alternative start from rest requires a compatible smooth inlet ramp and its own transient verification. The primary design uses the steady unladen field to avoid mixing inlet startup with particle response.

6.4 Geometry-aware representation

The fixed-width centerline mapping of the baseline cannot represent this variable-height domain without modification. A convenient mapping is

$$(x,\eta) \longmapsto (x,y) = (x, h(x)\eta), \qquad -1 \le \eta \le 1. \tag{6.3}$$

Its area Jacobian is $h(x)$. If the differential residual is evaluated in mapped coordinates, the chain rule must include this geometry dependence. For example,

$$\partial_x|_y = \partial_x|_\eta - \eta \frac{h'(x)}{h(x)}\partial_\eta, \qquad \partial_y = \frac{1}{h(x)}\partial_\eta. \tag{6.4}$$

Evaluating automatic derivatives directly with respect to physical coordinates is another option. Either route requires a manufactured derivative test before training a new flow problem.

Periodic Fourier features in the streamwise direction would impose the wrong structure for the proposed open domain. They must be replaced by nonperiodic coordinate features. A wall-distance factor can enforce the stationary wall velocity, but it must be consistent with the nonzero inlet condition and with the mapped derivatives [27]. A divergence-free streamfunction ansatz is a possible alternative, provided that its boundary and pressure conditions are verified. These are implementation choices to evaluate, rather than capabilities already established by the straight-channel model.

6.5 Gaussian support near the narrowing

The force-transfer rule must use quadrature over the fluid domain:

$$Z_i = \sum_{a\in\Omega_\delta} G(\boldsymbol{x}_a - \boldsymbol{q}_i)\Delta A_a, \qquad W_{ai} = \frac{G(\boldsymbol{x}_a - \boldsymbol{q}_i)}{Z_i}. \tag{6.5}$$

For mapped quadrature, ΔA_a includes the Jacobian from equation (6.3). The same points, weights, and normalizer must be used for interpolation and reaction-force deposition. Under that condition, the derivations of equations (5.8) and (5.9) remain valid on the new quadrature.

Figure 6.2 illustrates why normalization must follow the actual domain. Some Gaussian support intersects the wall region near the throat. Removing those points without updating Z_i would lose force. Renormalizing restores the integrated force, but changes the shape of the sampling operator, an issue related to coarse-graining choices [19]. Physical wall corrections [10, 11] and undisturbed-velocity recovery [17, 18] remain separate modeling requirements.

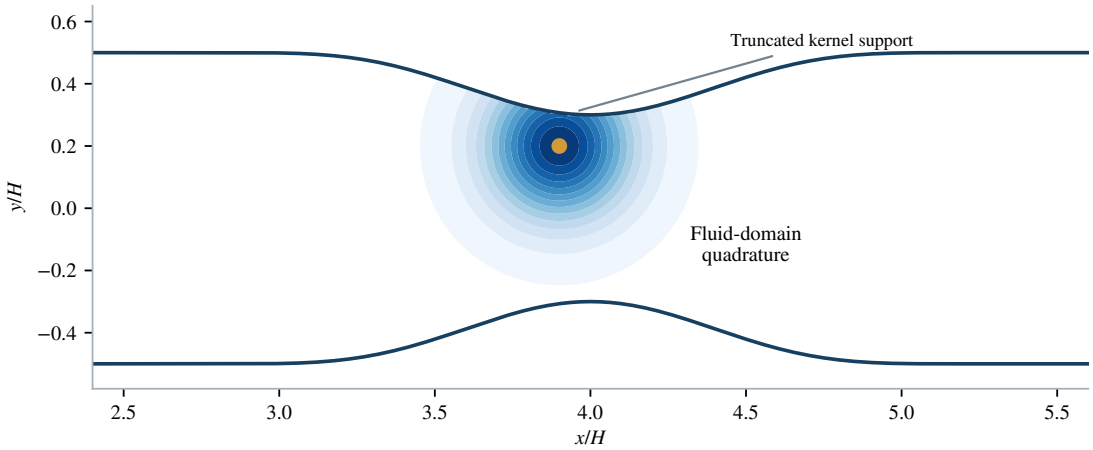


Figure 6.2. Illustration of a Gaussian centered near the upper throat wall. Shading is a prescribed kernel shape restricted to the proposed fluid domain, not a computed concentration or force field. The normalizer must use the same fluid-domain quadrature as both transfer operations.

Three ratios require explicit monitoring:

$$\frac{d}{H_t}, \qquad \frac{\sigma}{H_t}, \qquad \frac{\sigma}{\Delta_{\text{quad}}}, \tag{6.6}$$

where Δ_{quad} denotes a local quadrature-spacing measure. The first concerns physical confinement; the second concerns regularization relative to the gap; the third concerns the numerical resolution of that regularization. They address different questions and should be varied separately, with coarse-graining and transfer-resolution studies providing related context [19, 20]. If physical wall effects dominate, the present unbounded Stokes closure may need replacement before more accurate training is useful.

6.6 Particle and parcel wall conditions

An individual sphere has an admissible center region obtained by offsetting the wall by its radius. For a sloped wall, the offset is based on signed normal distance to the boundary, not only the vertical gap. A contact event must therefore use the actual geometry. The baseline's

irreversible sticking rule can be retained as a controlled modeling choice, but it must not be called a prediction of deposition physics.

A parcel introduces an additional issue: its center represents many particles. Stopping the entire represented population at the first center contact may create a strong and artificial capture rule. The proposed first study therefore focuses on carrier fields and freely transported dilute parcels whose centers remain inside the admissible region. A deposition extension should specify whether it resolves individual particles, splits parcels, or uses a separately justified population-level wall model. No capture efficiency is assigned before making that choice.

6.7 Controlled evaluation sequence

The extension uses staged verification consistent with the distinction between solving equations and establishing physical validity [31]. The following sequence is the study design adopted here.

1. Recover the straight, unladen open-channel solution at $\delta = 0$ with the new inlet and outlet conditions. Verify pressure gauge, wall velocity, and section flux before adding particles.
2. Solve the unladen constricted channel with an independent mesh-based method. Refine the throat and the upstream and downstream transition regions; monitor pressure drop and the complete velocity field.
3. Introduce individual one-way particles with identical initial conditions in the neural and reference fields. Compare position, common-time coverage, and endpoint class separately.
4. Add a single Gaussian parcel at low loading and verify force/work transfer on the irregular-domain quadrature. Compare the coupled fluid response with an independently discretized regularized model.
5. Vary severity and loading only after the earlier controls pass. Keep physical diameter, total represented mass, and kernel width explicit.

The initial proposed Reynolds numbers are 1, 5, and 20 at nominal height. These are design points rather than an observed regime map. Particle response times should be selected to retain an acceptably small slip Reynolds number at the throat. Mass-loading changes must be made through represented population while holding diameter fixed when the purpose is to isolate feedback. A separate diameter sweep changes both response time and confinement, and should be interpreted accordingly.

6.8 Observables and acceptance criteria

The primary observables are the section flux, throat velocity profile, pressure drop between fixed reference sections, and field error against the matched reference. For particles, residence time and endpoint class are added only after trajectory agreement is demonstrated. The reference sections must remain identical across severity values so that the pressure drop represents the same geometric interval.

At fixed flux, incompressibility gives the exact sectional mean relation

$$\overline{u}_x(x) = \frac{Q}{2h(x)}, \qquad \frac{\overline{u}_{x,t}}{\overline{u}_{x,0}} = \frac{1}{1-\delta}. \tag{6.7}$$

This is a geometric conservation implication, not a computed centerline velocity or a prediction of recirculation. A lubrication estimate, derived in chapter A, gives a useful additional check only when its small-slope and low-inertia assumptions apply.

Table 6.1. Proposed acceptance protocol for the constricted study. Limits must be fixed before interpreting the associated parameter sweep; these checks have not been executed.

Quantity	Proposed check	Interpretation
Unladen field	$E_u, E_p < 5\%$ against a matched reference	Carrier-field agreement
Reference refinement	Changes in Δp and field norms below 2%	Numerical sensitivity, not exact uncertainty
Section flux	Maximum relative variation below 2%	Integrated mass balance
Transfer	Force/work discrepancies consistent with roundoff	Algebraic operator consistency
Particle paths	RMS position below $0.05W$; common interval above 95%; all endpoint classes matched	Transport accuracy for stated initial conditions
Physical regime	Report maximum Re_p, d/H_t, local α and wall distance	Closure applicability; no universal threshold inferred

The error limits in table 6.1 are proposed numerical screens. In addition, residuals should be reported by component on a separate set of throat, near-wall, and bulk points. A global average must not conceal a concentration of error near the narrowing. The Gaussian width study should remain separate from spatial and temporal refinement, and multiple seeds should be included before drawing a quantitative trend.

6.9 Expected mechanisms and testable statements

Conservation at fixed flux predicts a larger section-mean speed as the gap narrows. The accompanying changes in particle transit depend on response time, local flow, and the retained force law. I treat changes in residence time, wall approach, and downstream recovery as hypotheses for the controlled study. Recirculation, enhanced deposition, or a particular pressure-loss factor cannot be inferred from the geometry alone.

At fixed imposed pressure difference, flux becomes an outcome instead. That experiment addresses a different question and must not be pooled with the fixed-flux design. Keeping these operating conditions separate is essential for a meaningful comparison of unblocked and constricted cases.

6.10 Chapter conclusions

The constricted-channel stage is defined at the study-design level through its geometry, boundary-condition choices, transfer consistency, and evaluation requirements. The outlet traction and its reference value must be fixed before executing the matched solvers. It extends the study in a controlled direction while preserving the distinction between mathematical consequences, proposed experiments, and completed numerical findings. Its next milestone is a verified unladen reference problem, followed by one-way transport and then two-way Gaussian feedback.

Chapter 7

Discussion

7.1 What the baseline establishes

The baseline demonstrates that a physics-informed field representation can agree closely with an independently discretized curved-channel solution for the specified steady, low-Reynolds-number problem. The velocity and pressure errors are below 2% and 1%, respectively, while the drag-network error is below 1%. The analytical plane-flow test provides an additional check that is independent of the curved-channel finite-element field. Reference-mesh sensitivity and section flux add evidence beyond a single parity plot.

The trajectory comparison evaluates an additional quantity beyond field agreement. Six of seven endpoint classes agree, while three of seven trajectories meet the complete position, common-time and endpoint criteria. A particle integrates the local field over time; force approximation, interpolation and event handling also affect its result. Their individual contributions require a controlled comparison. These findings motivate direct transport assessment alongside the carrier-field metrics.

The force network also raises a methodological question. Since the retained Stokes law is known analytically, its neural approximation is most useful as a controlled component of a broader learning framework. The present error does not show that the learned force is superior to the direct analytical formula. A future assessment should compare the two within identical carrier fields and identify whether the approximation adds cost, flexibility, or error.

7.2 What conservative transfer establishes

The shared Gaussian operator gives a stronger result of a different kind. The force and work identities follow from the normalization and the common quadrature. Their validity does not depend on a low training loss. This property removes one avoidable source of inconsistent interphase exchange and remains available when the carrier geometry changes, provided that the same fluid-domain quadrature is used consistently.

The identities do not guarantee the accuracy of the coupled solution. The carrier momen-

tum equation can retain a residual even when the sum of deposited forces is exact. In the evaluated coupled channel, the remaining momentum residual exceeds the study's bound in every tested parameter combination. The correct inference is that the operator design is verified while the quantitative coupled-channel prediction remains unresolved.

The successful reduced tests address important parts of the calculation: response-time dynamics, reciprocal momentum exchange, kernel integrals, and unsteady incompressible flow. Each isolates a necessary capability. Their combination is a verification chain, not a substitute for a matched independent two-way channel reference. This distinction is consistent with the separation between numerical verification and physical validation [30].

7.3 Optimization, residuals, and independent error

The completed calculations illustrate why optimization metrics and independent solution errors should be reported together. Residual normalization changes their numerical magnitude, and a weighted objective can favor some components over others. Reported raw residuals should therefore retain their equation convention. In particular, comparing the baseline momentum RMS directly with the Gaussian momentum RMS would ignore different scaling, geometry, and transient content.

Training difficulties in physics-informed learning are well documented [23–25]. They motivate focused improvements such as component scaling and more informative collocation. Residual-based adaptive sampling [37] and adaptive loss weighting [38] are options for a future controlled comparison. A further option is better temporal resolution of the response scale, including causality-aware training for the transient coupled problem [39]. These approaches were not part of the recorded runs. Their value must be assessed using the same independent checks and the original acceptance criteria.

7.4 Regularization and model validity

The Gaussian width controls how the represented force is distributed. It affects both the sampled fluid velocity and the carrier response. A smaller width is not automatically more physically accurate, especially if the transfer quadrature underresolves it. A larger width can smooth localized forcing while changing the regularized interaction. Width sensitivity must therefore be distinguished from convergence at fixed model parameters.

The undisturbed-velocity issue is equally important. The current parcels sample the disturbed fluid directly [17, 18]. A reduced Stokes-response test cannot resolve the resulting self-interaction question in a spatially coupled flow. Comparison with an independent solver of the same regularized equations would test numerical consistency; assessment against a suitable physical reference would additionally test the closure.

Confinement further separates numerical and physical accuracy. A model can converge accurately to equations that omit important wall effects. For that reason, the proposed constriction study records particle-to-gap ratio and local slip Reynolds number in addition to ordinary solver errors. If those quantities indicate a breakdown of the retained approximation, more training is not the appropriate physical correction.

7.5 Transferability across the three stages

Table 7.1 summarizes the actual scope of the findings. The baseline establishes a carrier-field comparison for one geometry and Reynolds number, with two seeds. The Gaussian tests establish specific reduced dynamics and transfer properties. The constriction stage provides a defined study design. None of these establishes a universal neural operator over channel shape, loading, and Reynolds number.

Table 7.1. Supported conclusions and their boundaries.

Stage	Supported statement	Remaining boundary
Baseline carrier	Analytical and independent numerical field agreement	One declared geometry/Re case; finite residuals
Baseline particles	Completed integrations and drag approximation	Six of seven matching endpoint classes; three complete path comparisons within criteria
Gaussian transfer	Discrete force/work consistency and component verification	Disturbed-velocity closure and coupled field error
Straight coupled channel	Executed diagnostic predictions	Momentum RMS 0.0407–0.3854 against a 0.03 criterion
Constriction	Defined geometry and evaluation protocol	No completed constricted-channel solution

The transferable parts are the nondimensional bookkeeping, reference matching, shared transfer, and evaluation logic. New physical and geometric claims require their own executed evidence. In particular, a parameter slot in a network does not establish learned parametric generalization, and a geometric illustration does not establish a flow solution.

7.6 Implications for the next computational study

The most informative next work is narrower than a large parameter sweep. First, the baseline trajectory limitation should be isolated using the same carrier field with exact and learned drag. Second, the Gaussian channel should be compared with an independent regularized solver at fixed width and well-resolved quadrature. Third, the unladen constricted problem should be solved before introducing moving populations.

This ordering separates carrier approximation, irregular-domain transfer, wall geometry, particle closure, and optimization. Establishing the controls in sequence allows each additional mechanism to be interpreted against a known numerical foundation.

7.7 Relationship to earlier methods

The present contribution is a continuous-time neural representation with shared, normalized force transfer and explicitly separated field and trajectory assessment. The transfer structure is related to Eulerian–Lagrangian methods [15, 20], while the fluid representation belongs to the velocity–pressure PINN framework [34]. Undisturbed-velocity recovery is a distinct modeling development addressed by Horwitz and Mani [17] and Ireland and Desjardins [18]. These relationships identify the design choices adopted here; the reported comparisons establish performance only for the stated tests and do not demonstrate general superiority over those earlier methods.

Chapter 8

Conclusions and Future Work

8.1 Principal conclusions

This thesis develops a staged framework for physics-informed fluid–particle transport. The progression begins with a one-way coupled particle problem, introduces Gaussian parcel reaction in a straight channel, and defines a smooth constriction for the next study. The completed evidence supports the following conclusions.

First, the baseline carrier network agrees with an independent finite-element solution of the stated curved-channel problem, with velocity and gauge-aligned pressure errors of 1.510% and 0.863%. Analytical plane-flow verification and reference-mesh sensitivity support this comparison. The normalized learned Stokes response has an independent error of 0.717%.

Second, carrier-field agreement does not by itself establish accurate particle transport. Six of seven baseline paths have matching endpoint classes, and three satisfy the combined endpoint, position, and time-coverage conditions. These comparisons define the quantitative scope of the particle results; deposition and wall-sensitive parameter trends require additional transport verification.

Third, a normalized Gaussian used for both interpolation and spreading preserves discrete force and work on the transfer quadrature. The resulting interphase drag power is nonpositive. These identities are algebraic properties of the discretization and do not imply that an imperfectly trained field satisfies every global balance.

Fourth, the reduced Gaussian-model tests recover Stokes response, uniform two-way relaxation, kernel integrals, and Taylor–Green decay with the reported errors and residuals. The accepted solution-observable errors in the listed relaxation and vortex tests are below 0.25%. The fully coupled straight-channel predictions are accompanied by momentum RMS values of 0.0407–0.3854 against the 0.03 reference criterion. They provide diagnostic evidence of the computed coupled response within this stated accuracy scope.

Finally, the constricted-channel extension has a consistent geometric and evaluation definition, including physical-coordinate derivatives, fluid-domain kernel normalization, and

independent throat-resolution checks. It has no completed predictive result in the present study.

8.2 Contributions

The study contributes a common evaluation framework connecting carrier fields, drag approximation, transfer consistency, and particle outcomes. It gives a dimensionally consistent planar sphere-parcel formulation and explicitly separates a Gaussian force distribution from a resolved solid body. It also defines which verified components can be retained when extending the model to a variable-height channel.

These contributions provide a practical basis for further computation without assigning unsupported accuracy to complex coupled predictions. The numerical record and portable manuscript sources preserve the arrays, criteria, and figure provenance needed to reassess the stated findings.

8.3 Priorities for future work

The immediate priority is to improve and independently verify the transport-sensitive parts of the calculation. For the baseline, this means near-wall field assessment and a direct exact-drag comparison. For Gaussian coupling, it means a matched independent regularized channel solution, transfer-quadrature refinement, temporal sampling refinement, and multiple training seeds at fixed physical parameters. Sampling and numerical-uncertainty methodologies provide a basis for planning those comparisons [31, 35, 37].

The constricted study should then begin with an unladen flow and a straight-channel boundary-condition control. One-way particles can be introduced after the carrier reference passes, followed by low-loading Gaussian feedback. Physical deposition or dynamic clogging requires a separately specified wall and population model. Experimental comparison will require matched geometry, operating conditions, particle properties, and observables before it can establish physical validity.

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Appendix A

Derivations and Transport Limits

A.1 Uniform two-way relaxation

For the uniform system in equation (5.17), define slip $w = v - u$. Adding the fluid and population-weighted particle equations gives

$$\frac{d}{dt}(u + \beta v) = 0. \tag{A.1}$$

Subtracting the equations gives

$$\dot{w} = -\frac{1+\beta}{\mathrm{St}} w, \qquad w(t) = w_0 \exp\left[-\frac{1+\beta}{\mathrm{St}} t\right]. \tag{A.2}$$

If $M_0 = u_0 + \beta v_0$ is the conserved momentum, the velocities are

$$u(t) = \frac{M_0 - \beta w(t)}{1+\beta}, \qquad v(t) = \frac{M_0 + w(t)}{1+\beta}. \tag{A.3}$$

For $u_0 = 0$, $v_0 = 1$ and $\beta = 0.25$, both velocities tend to 0.2. The total kinetic energy is

$$K = \frac{1}{2}u^2 + \frac{\beta}{2}v^2, \qquad \dot{K} = -\frac{\beta}{\mathrm{St}}(v-u)^2. \tag{A.4}$$

This gives the dissipative reference behavior of V02. It concerns a closed, uniform exchange system; the driven channel has additional work and boundary terms.

A.2 Adjoint transfer and kinetic energy

Let J denote the interpolation operator with entries $J_{ia} = W_{ai}\Delta A_a$. With quadrature mass matrix $M_A = \mathrm{diag}(\Delta A_a)$, the spreading operator is $-M_A^{-1}J^{\mathsf{T}}$ when the parcel force vector stores integrated forces. Thus

$$\boldsymbol{U} = J\boldsymbol{u}, \qquad \boldsymbol{S} = -M_A^{-1}J^{\mathsf{T}}\boldsymbol{F}. \tag{A.5}$$

The fluid work is $\boldsymbol{u}^{\mathsf{T}} M_A \boldsymbol{S} = -(J\boldsymbol{u})^{\mathsf{T}} \boldsymbol{F}$. Adding parcel drag work produces equation (5.10). This matrix statement makes explicit why interpolation and spreading must use the same quadrature and weights.

For a collision-free driven channel with no-slip stationary walls and periodic streamwise faces, a formally exact solution obeys

$$\begin{aligned}\frac{d}{dt}\left[\frac{1}{2}\int_\Omega |\boldsymbol{u}|^2\,dA + \sum_i \frac{m_i}{2}|\boldsymbol{v}_i|^2\right] &= \int_\Omega \boldsymbol{u}\cdot\boldsymbol{f}_d\,dA + \sum_i m_i \boldsymbol{v}_i\cdot\boldsymbol{g}_{\text{eff}} \\ &\quad - \frac{1}{\mathrm{Re}_H}\int_\Omega |\nabla\boldsymbol{u}|^2\,dA - \sum_i \frac{m_i}{\mathrm{St}_i}|\boldsymbol{U}_i - \boldsymbol{v}_i|^2.\end{aligned} \tag{A.6}$$

Pressure and periodic-face contributions cancel under the stated conditions. The wall velocity is zero, so stationary walls do no work. Quadrature and neural equation residuals can create a nonzero discrepancy when this identity is evaluated numerically. Wall tractions still enter the momentum budget; they must not be omitted merely because wall work vanishes in the energy budget.

A.2.1 Definitions of the plotted balance discrepancies

The plotted diagnostics use the same midpoint weights $\Delta A = 3/256$ as the transfer. Define the discrete total momentum and energy by

$$\boldsymbol{M} = \sum_a \boldsymbol{u}_a \Delta A + \sum_i m_i \boldsymbol{v}_i, \qquad K = \frac{1}{2}\sum_a |\boldsymbol{u}_a|^2 \Delta A + \frac{1}{2}\sum_i m_i |\boldsymbol{v}_i|^2. \tag{A.7}$$

The signed energy discrepancy in figure 5.10 is

$$r_E = \frac{dK}{dt} + D_\nu + D_{\text{drag}} - W_{\text{drive}} - W_g, \tag{A.8}$$

with

$$D_\nu = \frac{1}{\mathrm{Re}_H}\sum_a |\nabla\boldsymbol{u}_a|^2 \Delta A, \qquad D_{\text{drag}} = \sum_i \frac{m_i}{\mathrm{St}_i}|\boldsymbol{U}_i - \boldsymbol{v}_i|^2, \tag{A.9}$$

$$W_{\text{drive}} = \sum_a \boldsymbol{u}_a \cdot \boldsymbol{f}_d \Delta A, \qquad W_g = \sum_i m_i \boldsymbol{v}_i \cdot \boldsymbol{g}_{\text{eff}}. \tag{A.10}$$

For momentum, the plotted quantity is the Euclidean norm of

$$\boldsymbol{r}_M = \frac{d\boldsymbol{M}}{dt} - \boldsymbol{T}_\nu - \boldsymbol{F}_p - |\Omega|\boldsymbol{f}_d - \sum_i m_i \boldsymbol{g}_{\text{eff}}. \tag{A.11}$$

Here $\boldsymbol{T}_\nu = \mathrm{Re}_H^{-1}\int_{\partial\Omega_w} \partial_n \boldsymbol{u}\,ds$ is the viscous wall-flux contribution for the Laplacian form of the momentum equation, and $\boldsymbol{F}_p = -\int_{\partial\Omega_w} p\boldsymbol{n}\,ds$ is wall-pressure force. Opposite periodic-face

terms cancel. Wall integrals use midpoint sampling along the walls. Internal drag cancels between the two phases.

The time derivatives of K and $\boldsymbol{M}$ are obtained by automatic differentiation of the neural fields and parcel states at fixed spatial quadrature points. They are instantaneous derivative discrepancies, not cumulative integration errors. The plots contain 41 stored times over $0 \leq t \leq 1$. All terms use the Gaussian channel's nondimensional inertial convention; their absolute magnitudes are not percentages. The separate 0.03 screen concerns held-out differential-residual RMS, not these integrated balance curves.

A.3 A limiting pressure-drop estimate

For slowly varying plane walls and negligible inertia, the local lubrication approximation gives the dimensional velocity

$$u(x,y) = -\frac{1}{2\mu}\frac{dp}{dx}[h(x)^2 - y^2]. \tag{A.12}$$

Integrating over $-h \leq y \leq h$ yields

$$Q = -\frac{2h(x)^3}{3\mu}\frac{dp}{dx}, \qquad \Delta p = \frac{3\mu Q}{2}\int_{x_a}^{x_b}\frac{dx}{h(x)^3}. \tag{A.13}$$

For constant half-height h_0, this reduces to the plane Poiseuille pressure drop $3\mu Q(x_b - x_a)/(2h_0^3)$. At fixed flux, the normalized hydraulic resistance estimate is

$$\frac{\Delta p}{\Delta p_0} = \frac{h_0^3}{x_b - x_a}\int_{x_a}^{x_b}\frac{dx}{h(x)^3}. \tag{A.14}$$

This is a limiting analytical check, not a simulation result for the proposed severity values. Its reliability requires sufficiently small wall slope and inertial effects; a full Navier–Stokes solution is needed when those assumptions fail [40].

A.4 Baseline trajectory criteria

Table A.1 retains the seven baseline comparisons associated with the field results in chapter 4. RMS position discrepancy is normalized by $W = 0.3$. The current numerical limits require discrepancy below 5%, common-time coverage above 95%, and agreement of every endpoint class. Some individual rows satisfy those conditions, but the complete set does not. The trajectories are consequently not used as validated deposition evidence.

Table A.1. Complete baseline particle comparison; no initial position is removed to improve the aggregate outcome.

Initial η	RMS position/W (%)	Common time (%)	Same endpoint class
-0.650	2.67	98.86	Yes
-0.433	2.30	99.33	Yes
-0.217	4.30	99.27	Yes
0.000	7.83	98.59	Yes
0.217	16.09	96.19	Yes
0.433	34.97	87.39	Yes
0.650	3.29	73.55	No

A.5 Planar mass convention

For a physical sphere of diameter d, mass is $\rho_p \pi d^3/6$. A parcel containing N_i such spheres has mass $N_i \rho_p \pi d^3/6$. Dividing by $\rho_f H^2 B$ gives

$$\frac{N_i \rho_p \pi d^3/6}{\rho_f H^2 B} = \frac{N_i R \pi (d/H)^3}{6(B/H)}, \tag{A.15}$$

which is equation (3.9). The dimensional reaction source in a planar fluid equation has force divided by depth and area. This is why a two-dimensional Gaussian normalization uses area rather than volume. The reference depth records the physical interpretation without introducing a resolved spanwise flow field.

Appendix B

Additional Preliminary Results and Validation

The nineteen figures in the core study are supplemented here by eight additional figures. These examine integrator accuracy, the isolated drag approximation, optimization histories, independent verification errors, and parcel bookkeeping. A separate resolved-cylinder calculation supplies a preliminary comparison with published equilibrium-rotation values. Each comparison retains its own physical model, reference, and scope.

B.1 Independent Stokes-relaxation integrator check

The baseline particle integrator is checked in a uniform carrier field against $v(t) = 1 - \exp(-t/\tau)$, with $v(0) = 0$ and $\tau = 0.0347222$. The stored Radau integration uses relative and absolute tolerances 10^{-9} and 10^{-11}. Across all 150 saved times over $0 \le t/\tau \le 6$, the largest absolute velocity discrepancy is 1.5351×10^{-10}. This is an independent check of the ODE integration and prescribed Stokes acceleration.

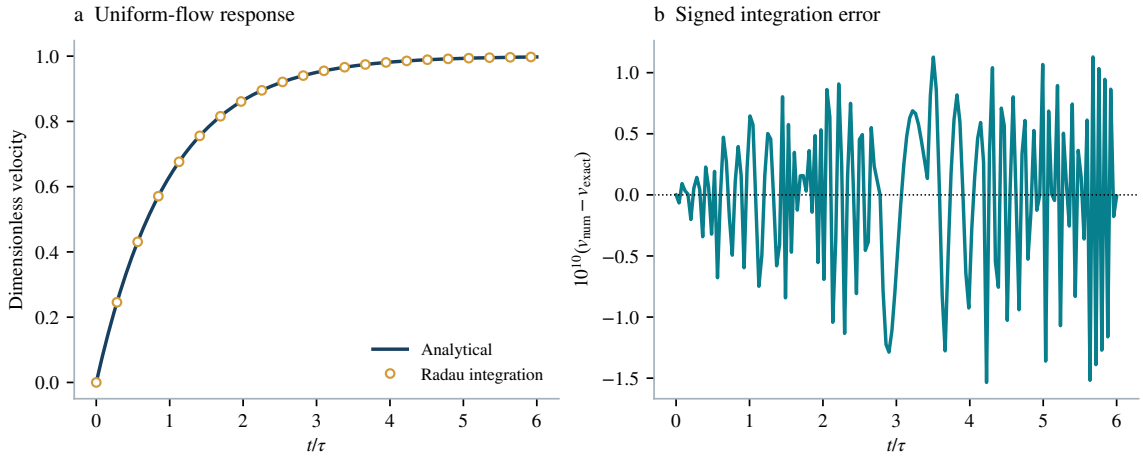


Figure B.1. Uniform-flow relaxation from the baseline record. The left panel compares the stored numerical solution with the analytical response; markers are subsampled for readability. The right panel retains the signed error at every saved time. This test concerns the ODE integrator and exact Stokes law, separately from neural-field and learned-drag errors.

B.2 Drag approximation with the carrier field held fixed

The recorded isolation study compares PINN–ForceNet trajectories with trajectories using analytical Stokes drag in the same PINN carrier field. All seven baseline initial conditions are retained. Their endpoint classes agree, common-time coverage is at least 99.8619%, and the largest RMS position discrepancy divided by channel width is 0.4279%. Holding the carrier field fixed isolates the effect of the drag approximation within this comparison. The independent PINN–FEM trajectory assessment in table A.1 addresses a broader error.

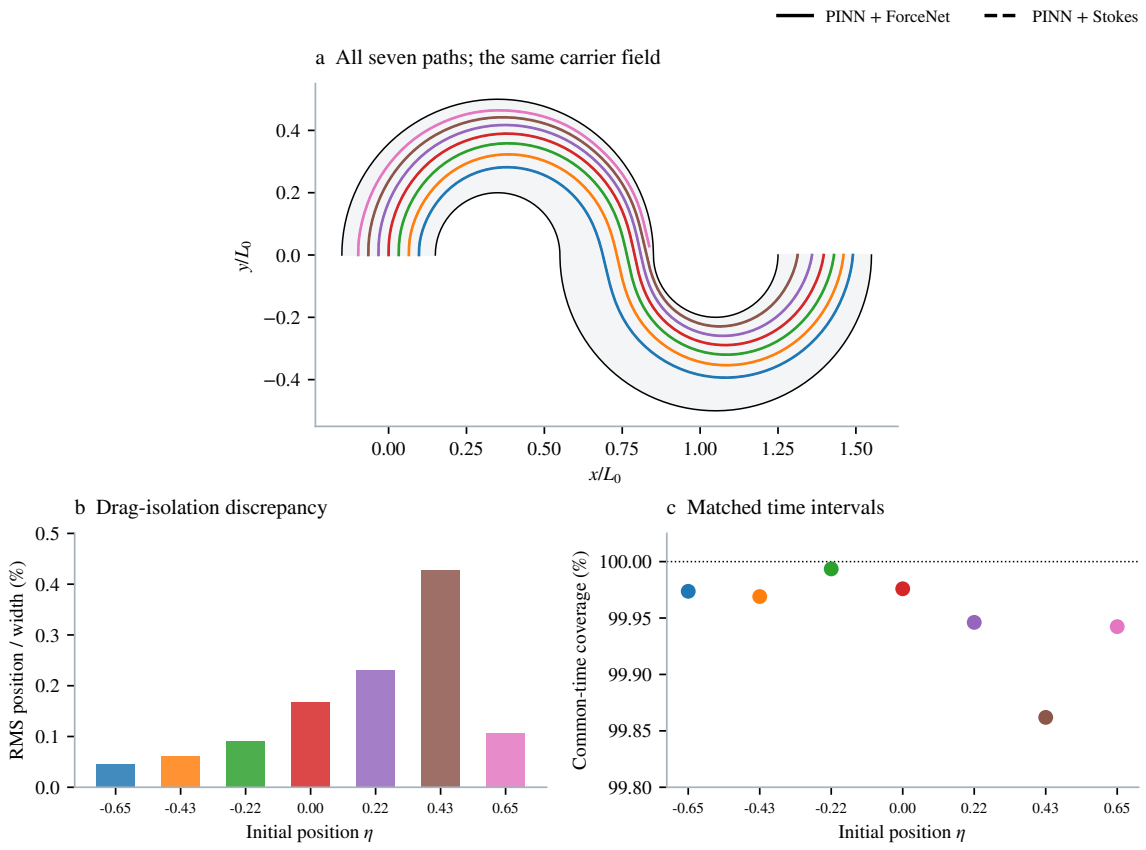


Figure B.2. Learned and analytical drag in the same carrier field, with matching colors for each initial position. The spatial panel displays the stored paths; the lower panels show RMS position discrepancy normalized by $W = 0.3$ and the coverage of the shared physical-time interval. Every initial position in the isolation record is included. The comparison supports the drag approximation within the fixed learned field.

B.3 Recorded optimization histories

The archived histories retain the optimization of both baseline flow seeds and the ForceNet training and validation losses. Figure B.3 shows all logged records on logarithmic axes, without smoothing or replacing the original sequences. The flow monitor falls from its initial scale to 2.75×10^{-3} and 3.08×10^{-3} for the two seeds. Independent field and drag

comparisons provide the accuracy assessment alongside these optimization diagnostics.

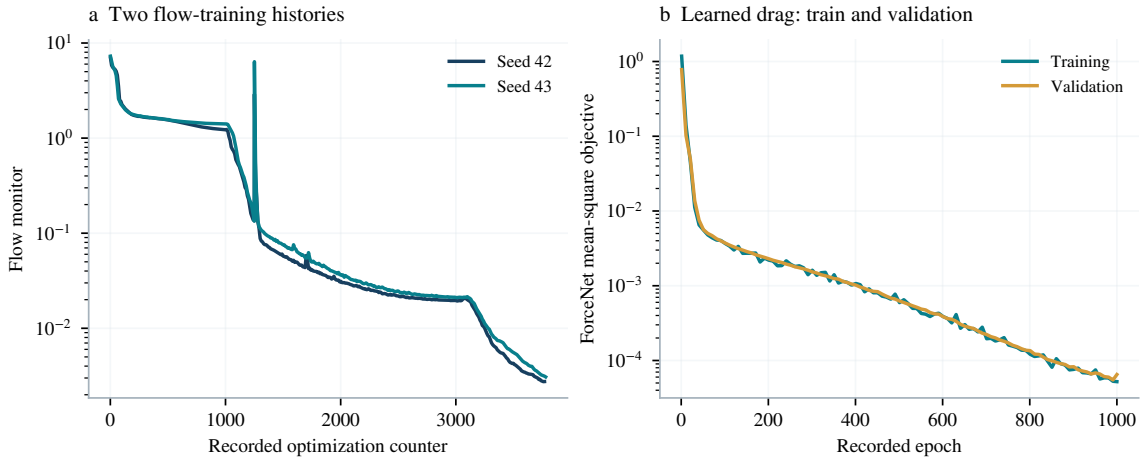


Figure B.3. Recorded flow-monitor histories for seeds 42 and 43, and the ForceNet training and validation objectives. The horizontal coordinate is the stored epoch or optimization counter; it is not elapsed runtime. Curves use every logged sample. The loss conventions of the flow and drag models differ, so their vertical magnitudes are interpreted separately.

B.4 Pointwise errors in the Gaussian verification problems

The reduced Stokes and uniform two-way tests supply full temporal arrays, while the Taylor–Green comparison supplies 768 independently evaluated space–time points. Plotting prediction minus reference makes the error structure visible in addition to the solution comparisons in figures 5.2, 5.3 and 5.5. All supplied comparison samples are retained in figure B.4. These are dimensionless absolute discrepancies in the indicated observables.

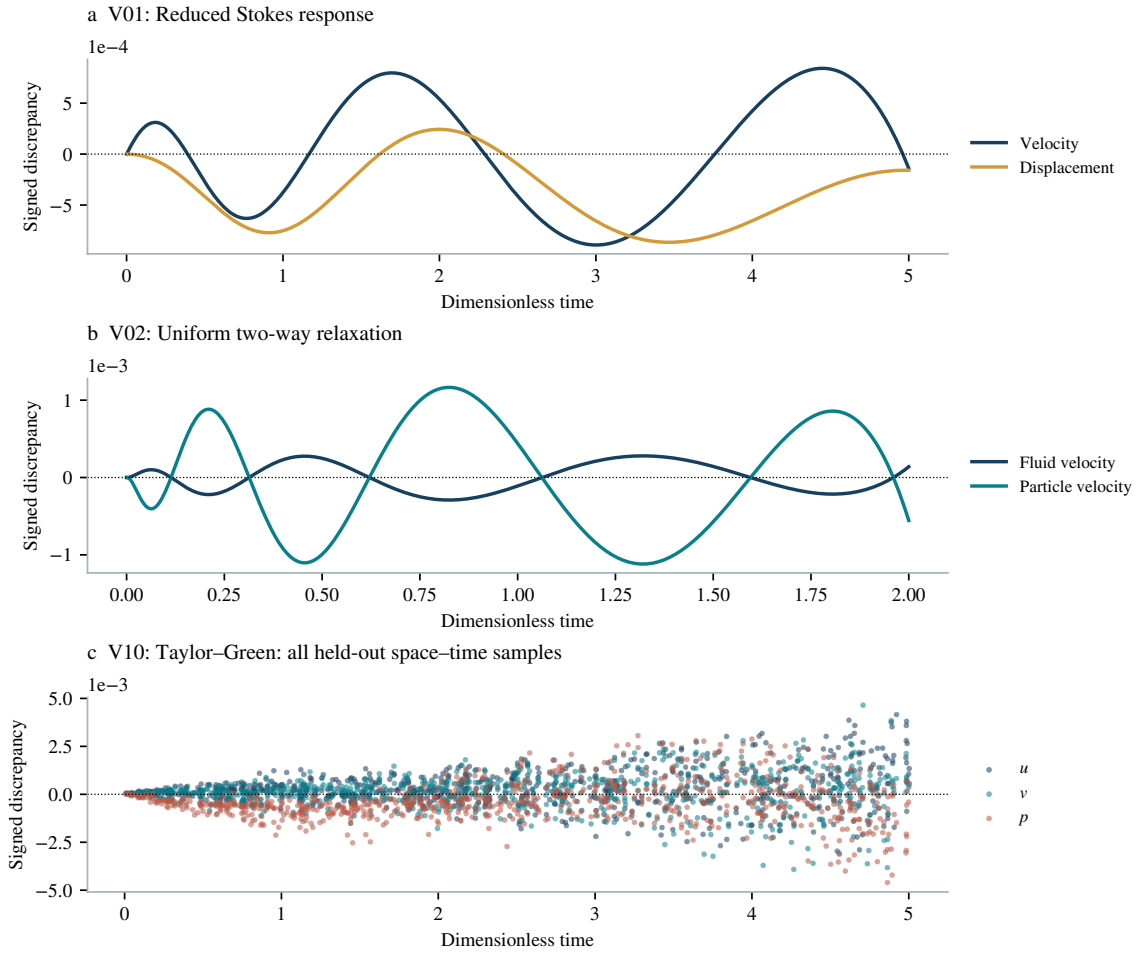


Figure B.4. Signed prediction-minus-reference discrepancies for the reduced Stokes response, uniform two-way relaxation, and Taylor–Green carrier test. The Taylor–Green panel plots each saved error against its sampled time; the spatial coordinates vary between samples. Pressure uses the common gauge of the original verification. These component tests retain the initial constraints and independent-reference scope described in chapter 5.

B.5 Population-preserving parcel splitting

Test V09 divides a fixed population of 10,000 particles into 10, 50, or 100 coincident parcels with adjusted multiplicities. The deposited force density differs by at most 2.6645×10^{-15} from the ten-parcel reference. Figure B.5 shows both the preserved population and the associated floating-point-scale discrepancy. This is a linearity and population-bookkeeping check at coincident positions; spatial and statistical refinement have separate meanings.

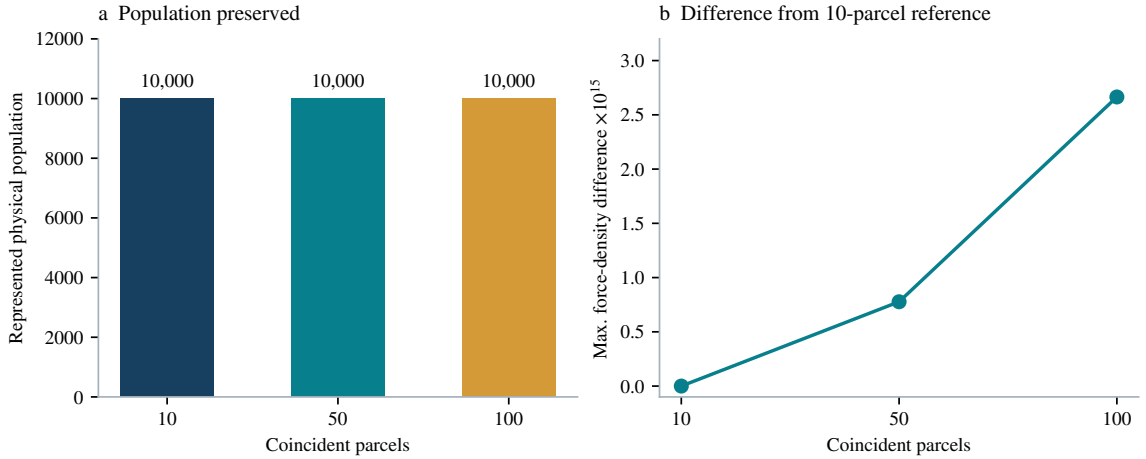


Figure B.5. V09 parcel-splitting check. The physical population remains 10,000 in all three representations. Force-density differences are measured relative to the ten-parcel representation and are shown on a linear scale in units of 10^{-15}, including the exact zero of that reference. The coincident positions and fixed total population define the scope of this algebraic transfer test.

B.6 Standalone impact-algebra checks

The optional interaction framework includes separately tested binary impact and wall-restitution operations. Tests V05 and V06 use equal unit masses, with restitution coefficients 0.8 and 1.0. Test V07 uses a stationary wall and restitution 0.6. Their stored outgoing velocities agree with the corresponding impulse and reflection formulas. Figure B.6 compares the before/after velocity components. These are impulsive component checks, distinct from a time-resolved soft-contact or collision-coupled channel calculation.

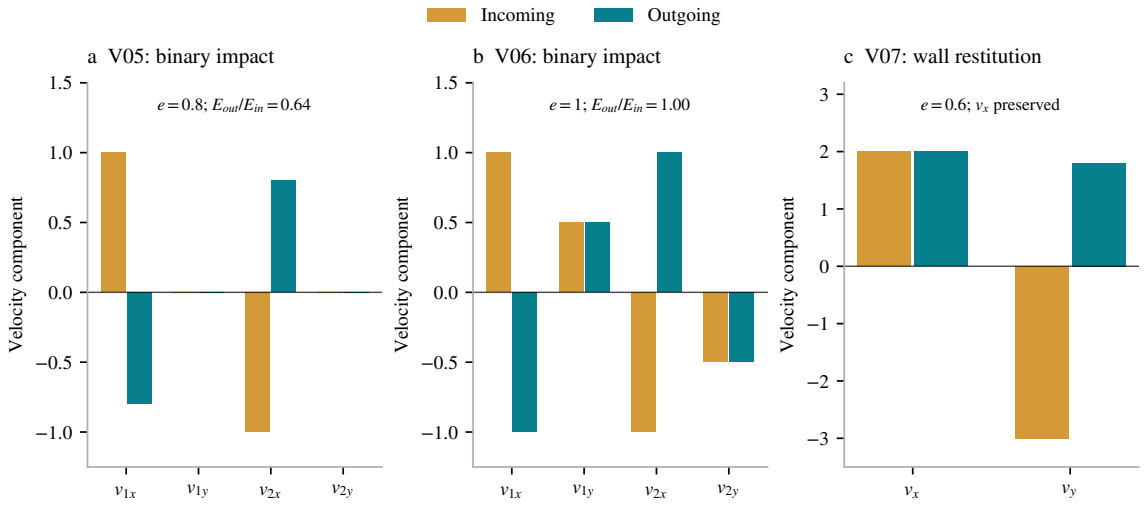


Figure B.6. Velocity-component comparisons for V05, V06 and V07. Paired bars show incoming and recorded outgoing values. The inelastic equal-mass impact retains 64% of its initial kinetic energy; the elastic oblique impact preserves kinetic energy; the wall operation preserves tangential velocity and reverses the normal component with restitution 0.6. All values come from the recorded primitive tests.

B.7 Preliminary resolved-cylinder Couette comparison

A separate extension resolves the surface of a circular cylinder in planar Couette flow. The domain is $[0,2]\times[0,1]$, the diameter is 0.25, and the prescribed center is $(1,0.5)$. In the translating frame, the wall velocities are $(-0.5,0)$ and $(0.5,0)$, with periodicity in x. The viscosity is $1/(64\mathrm{Re}_p)$ for the two evaluated conditions $\mathrm{Re}_p = 1$ and 2. The cylinder angular velocity is determined by a torque-free condition. Published finest-grid equilibrium values provide the scalar reference [41].

The neural calculation uses independently generated FEM velocity anchors; it is therefore a FEM-assisted PINN. The published angular velocities are reserved for comparison. Their magnitudes are 0.4608 and 0.4281; the corresponding PINN values are 0.459740 and 0.427275. Relative differences are 0.2300% and 0.1926%. The independent FEM values differ from the published values by 0.1347% and 0.1787%. The two-mesh changes in FEM angular velocity are 0.00567% and 0.00787%.

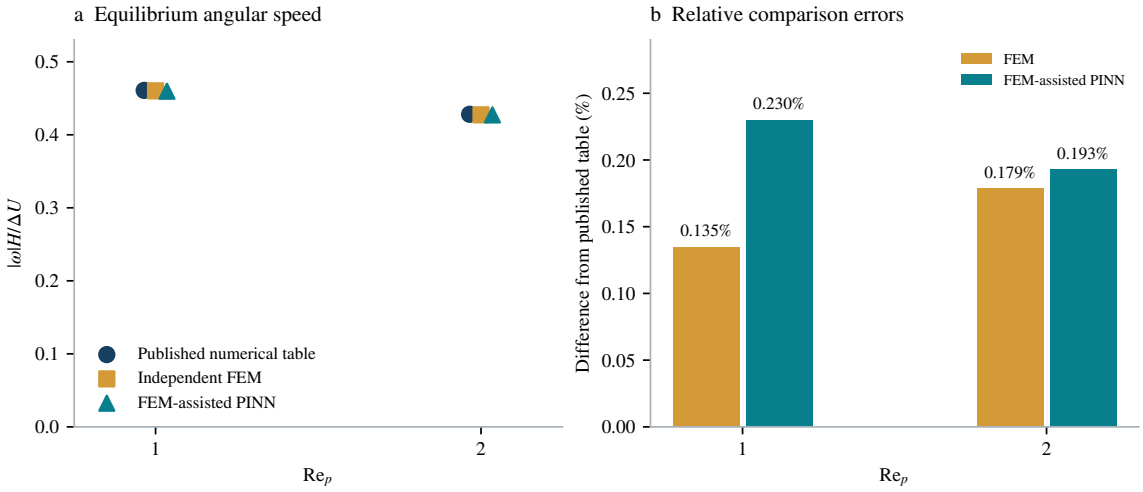


Figure B.7. Published numerical equilibrium values, independent FEM, and FEM-assisted PINN at the two evaluated particle Reynolds numbers. The second panel shows the relative angular-velocity differences against the published table. Both operating conditions in the completed comparison are included; small horizontal marker offsets distinguish the three methods. The comparison concerns equilibrium rotation at a prescribed symmetry-plane center; the reference is a numerical study.

B.8 Resolved-cylinder fields and independent diagnostics

The stored comparison fields supply the velocity distributions in figure B.8. The circular body is masked using the stored fluid sampling locations. For the independent assessment samples, PINN–FEM relative velocity errors are 0.1682% and 0.2038%. The momentum RMS values are 0.003322 and 0.002612, and the continuity RMS values are 0.001175 and 0.001195. Normalized torque residual magnitudes are 8.72×10^{-6} and 5.73×10^{-6}. The exact boundary construction gives zero recorded wall-velocity error and surface errors below 3.5×10^{-16}.

These results support the stated equilibrium comparison for a resolved cylinder with FEM-assisted training. The Gaussian point-parcel model and the proposed constricted channel retain their own definitions and evaluation requirements. The stored steady equilibrium fields also have a different scope from a transient migration trajectory.

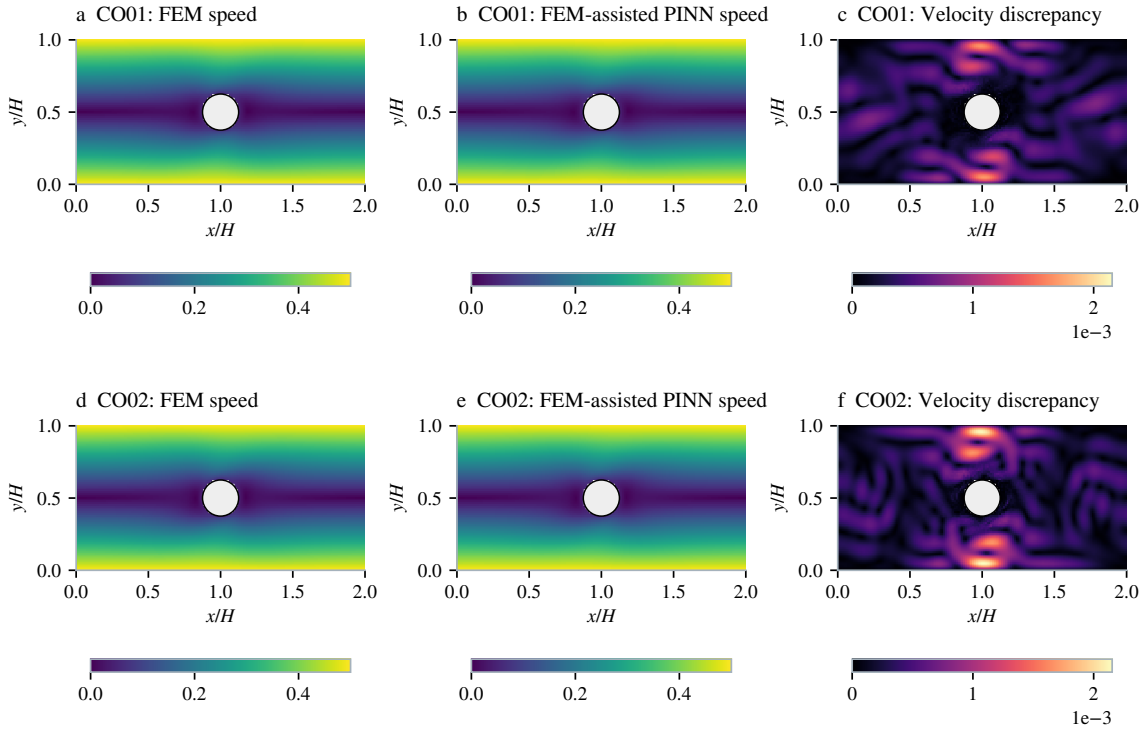


Figure B.8. Stored FEM and FEM-assisted PINN speed fields for CO01 and CO02, with the norm of their local velocity difference. A common speed scale is used in all four field panels and a common discrepancy scale in both error panels. The circular solid is excluded from the plotted fluid domain. These are the recorded visualization-grid samples; the independent evaluation errors quoted in the text use their separate assessment set.